Ontology Tradcraft

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CEO, *Acacia Knowledge Systems Inc.*

Outline

- How Field are Born
- New Era, New Tradecraft
- Tradecraft as Recipe
- Tradecraft as Encoding

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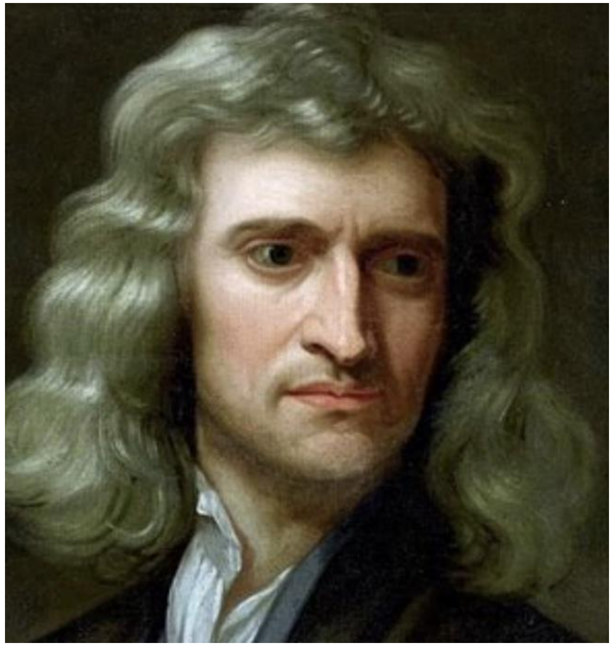
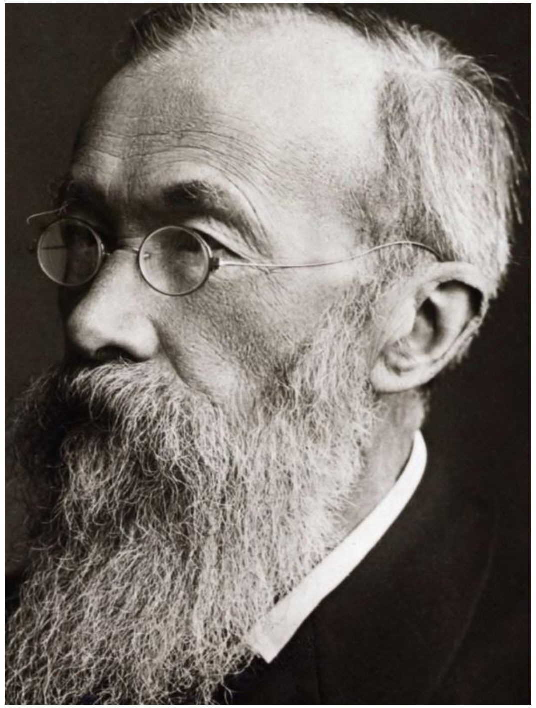
Philosophy is at heart the **study of methods**; philosophers seek out methods for answering important questions

Once consensus has been reached over methods, they **form the basis of a scientific discipline**

Garden of Forking Paths

- Philosophy is a garden in which scientific disciplines are grown





Garden of Forking Paths

- Philosophy is a garden in which scientific disciplines are grown
- Ontology engineering – despite existing as a field for over 20 years – has yet to fully emerge as a scientific discipline

Garden of Forking Paths

- Philosophy is a garden in which scientific disciplines are grown
- Ontology engineering – despite existing as a field for over 20 years – has yet to fully emerge as a scientific discipline
- Philosophers turned engineers and engineers turned philosophers **do not** a discipline make
- We need **consensus on methods**

WHAT DISTINGUISHES ONTOLOGY ENGINEERING FROM NEARBY DISCIPLINES?

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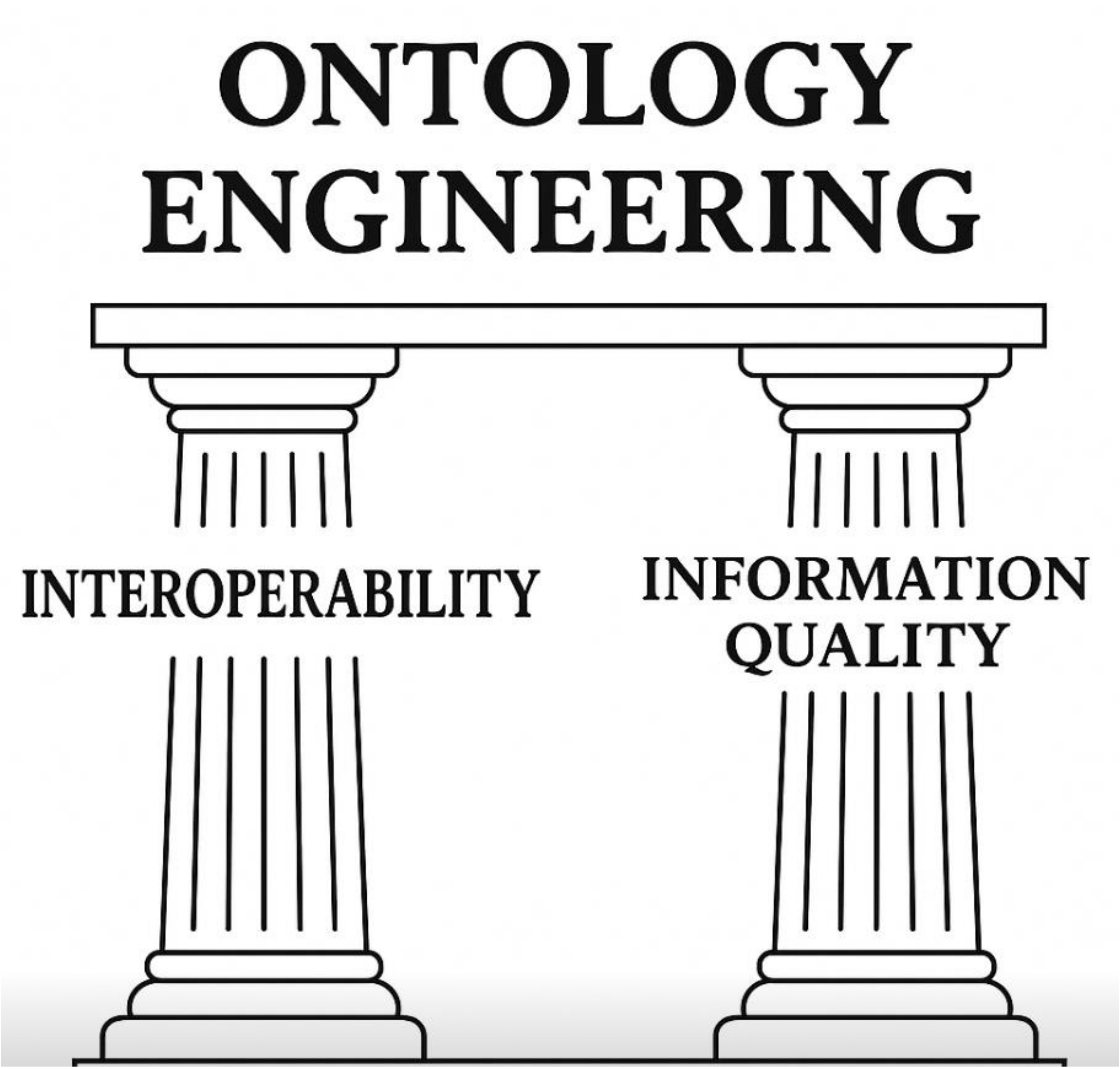
ONTOLOGY
ENGINEERING
INTEROPERABILITY
INFORMATION
QUALITY


ONTOLOGY
ENGINEERING
INTEROPERABILITY
INFORMATION
QUALITY

Promise of Ontology Engineering

- Ontologies are formally well-defined machine-interpretable controlled vocabularies designed to represent entities and logical relationships among them
- Ontologies make **explicit** the **implicit** meanings buried in datasets, by using basic principles of formal logic
- Ontologies provide a **semantic layer** to connect information silos

Interoperability Strategies

- For the sake of argument, let us call "interoperability strategies" those strategies that mitigate or eliminate information silos

- Interoperability strategies may be divided along at least three axes

Human-Human
Human-Machine
Machine-Machine

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Central Conceit

- Concerns the alignment of understanding between **human agents**
- Often pursued through the **creation of** shared vocabularies, definitions, standards, and practices, perhaps facilitated by education, legislation, consensus-building, or social conventions

Any two speakers of a given natural language share some underlying formal structure

Interoperability Strategies

- For the sake of argument, let us call "interoperability strategies" those strategies that mitigate or eliminate information silos

- Interoperability strategies may be divided along at least three axes

Human-Human
Human-Machine
Machine-Machine

Central Conceit

- Concerns the **capability of computational systems** to interpret and act upon human-understandable descriptions of the world

- Promoting human-machine interoperability requires translating domain knowledge into formal representations that machines can reason over

Any two heterogeneous encodings share some underlying formal structure

Interoperability Strategies

- For the sake of argument, let us call "interoperability strategies" those strategies that mitigate or eliminate information silos

- Interoperability strategies may be divided along at least three axes

Human-Human
Human-Machine
Machine-Machine

Central Conceit

- Concerns the extent to which **independent computing systems can exchange, interpret, and process data** without human intervention

- Requires robust data structuring, shared semantics, exchange protocols

Any two heterogeneous information systems share some underlying formal structure


ONTOLOGY
ENGINEERING
INTEROPERABILITY
INFORMATION
QUALITY

Connection between Pillars

- Information Quality: Fitness of information for its intended use
- Interoperability is often pursued without reflection on how information quality will or may be impacted in applications
- Information quality is often pursued without reflection on how interoperability will be obtained

Connection between Pillars

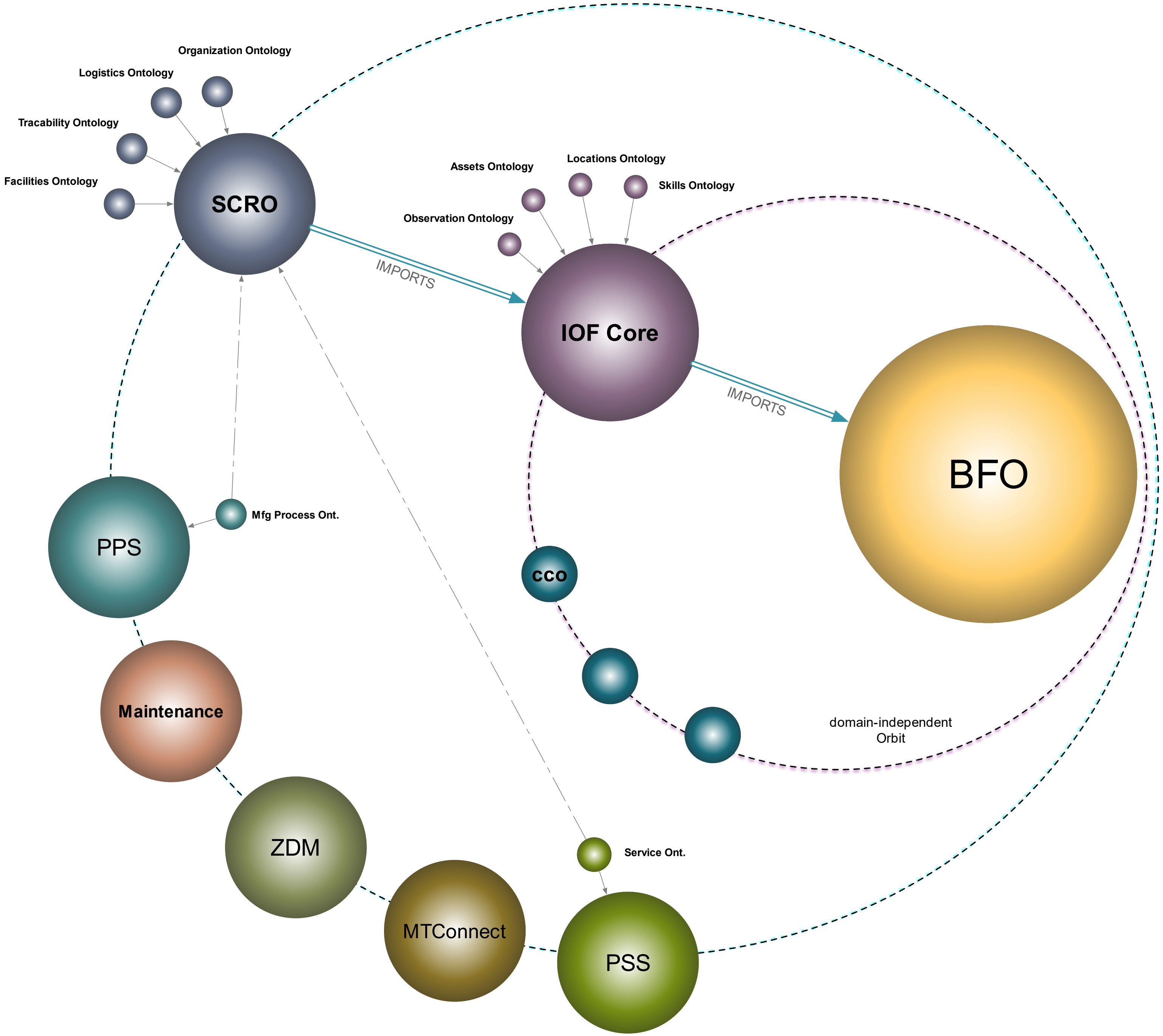
- Too much **emphasis on interoperability** leads to ontology artifacts that lack the the rich axiomatization needed to leverage tools to support information quality improvement

- Too much **emphasis on the information quality** results in information silos, unable to leverage common semantics in computing systems to promote interoperability

Best Practices

- Pursuing **interoperability** and **information quality** simultaneously can be pursued by adopting common best practices in our community

- Many of which are shared with the software development community, such as having a **top-level ontology**, adopting **modularization**, and so on



Organization Ontology
Logistics Ontology
Tracability Ontology
Facilities Ontology
SCRO
Assets Ontology
Locations Ontology
Skills Ontology
Observation Ontology
IMPORTS
IOF Core
IMPORTS
BFO
PPS
Mfg Process Ont.
CCO
Maintenance
domain-independent
Orbit
ZDM
MTConnect
Service Ont.
PSS

KNOWN KNOWNS

“things that we’re aware that we know”

KNOWN UNKNOWNS

“things that we’re aware that we **don’t** know”

UNKNOWN KNOWNS

“things that we’re **unaware** that we know”

UNKNOWN UNKNOWNS

“things that we’re **unaware** of and **don’t** know”

INTEROPERABILITY

KNOWN KNOWNS

"things that we're aware that we know"

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INFORMATION QUALITY

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UNKNOWN
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INFORMATION QUALITY

Systematic Disambiguation

- Ontology engineering is a discipline concerned with **systematic disambiguation**

- "Disambiguate" does not mean find and represent a **privileged** interpretation of some phenomenon

- It means find and represent the **logical space of interpretations** of some or some related phenomena

Disambiguation

- **Type** vs an **instance** of that type, e.g. the instance Algeria vs country
- **Information** vs what that information **is about**, e.g. occupation code vs a holder of an occupation
- **Material** vs **immaterial** things, e.g. a given river vs the site where the river used to flow
- **Processes** vs **product**, e.g. ontology engineering vs ontology produced

Leibniz's Law

IF *X* HAS A PROPERTY THAT *Y* DOES NOT HAVE, THEN *X* CANNOT BE IDENTICAL TO *Y*

An occupation code is a string of symbols; an occupation holder is a not

A river consists of flowing water; the site where a river flowed need not

Systematic Disambiguation

- Ontology engineering is a discipline concerned with **systematic disambiguation**
- "Systematic" means disambiguate the **logical space of interpretations** of some related phenomena in a principled, logically exhaustive manner
- It does not mean **boil the ocean**; more like **boil rivers draining into the ocean**

Finer Point

- I began with the question "what distinguishes ontology engineering from nearby disciplines"

- The answer is our north star, it will guide our consensus, it is the rallying cry of the new era:

Ontology engineering is focused on constructing machine-interpretable artifacts designed to systematically disambiguate information, improve information quality, and facilitate information interoperability

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SUPPOSE YOU'RE GIVEN THE TASK OF REPRESENTING SOME DOMAIN

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IN THE INTEREST OF USING MACHINES TO ANSWER IMPORTANT QUESTIONS ABOUT IT

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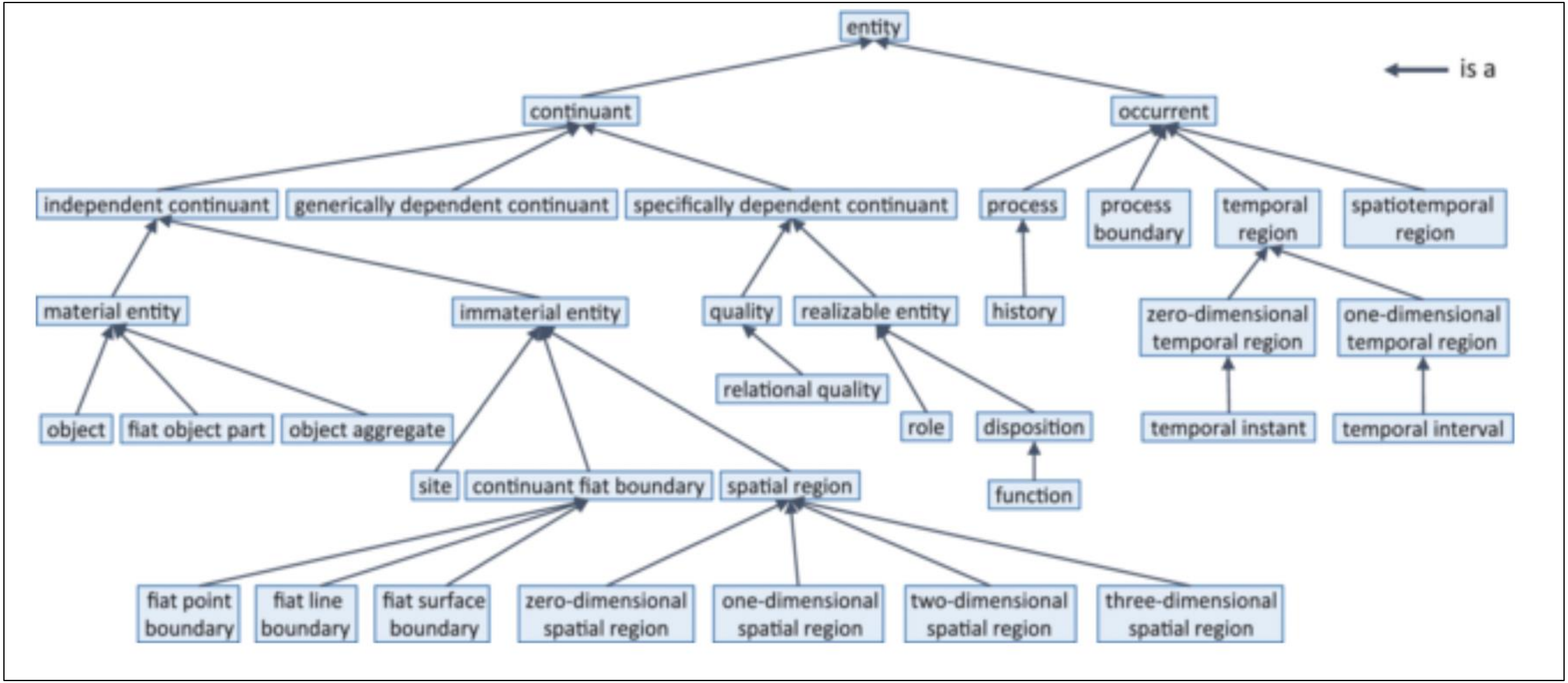
EASY, JUST BUILD A BFO MODEL!



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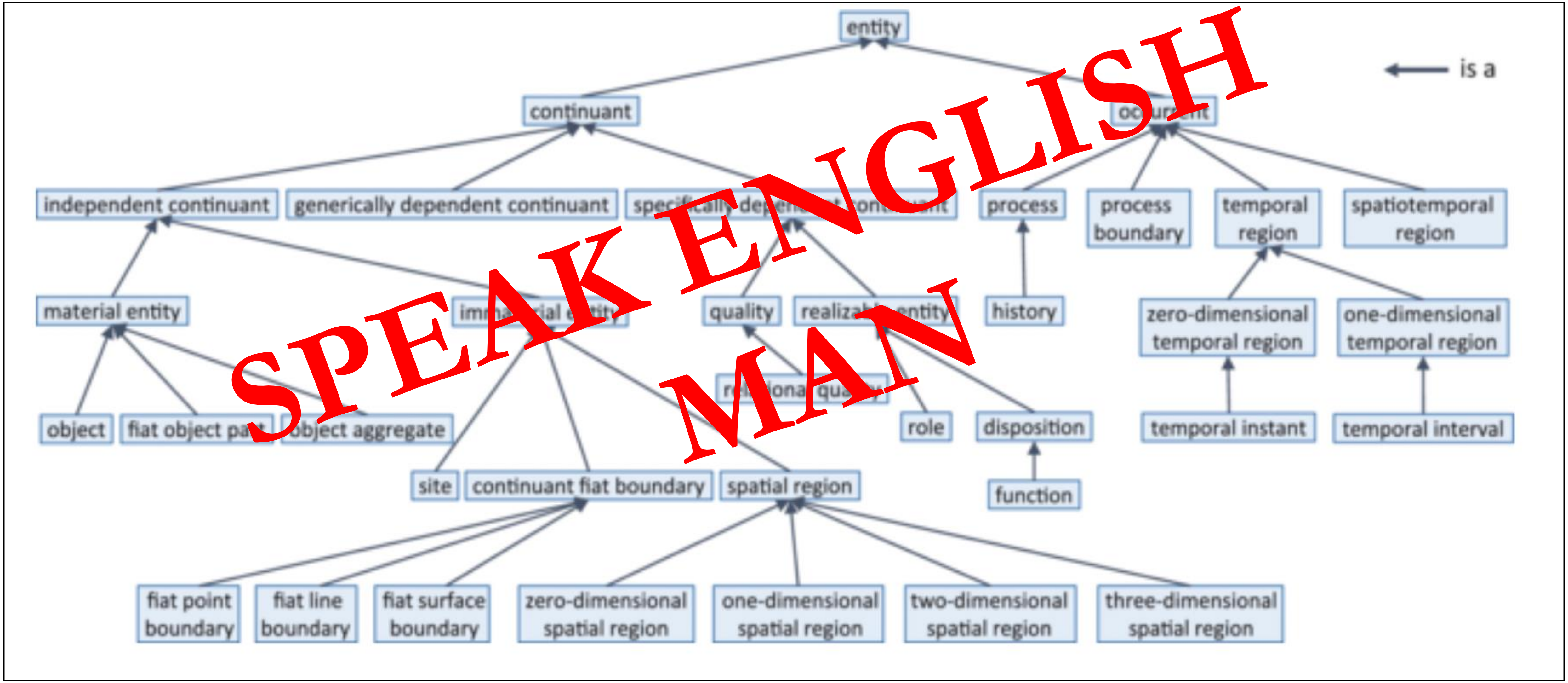
occurrent
independent continuant
generically dependent continuant
specifically dependent continuant
process
process boundary
temporal region
spatiotemporal region
material entity
immaterial entity
quality
realizable entity
history
zero-dimensional temporal region
one-dimensional temporal region
relational quality
object
fiat object part
object aggregate
role
disposition
temporal instant
temporal interval
site
continuant fiat boundary
spatial region
function
fiat point boundary
fiat line boundary
fiat surface boundary
zero-dimensional spatial region
one-dimensional spatial region
two-dimensional spatial region
three-dimensional spatial region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SPEAK ENGLISH
MAN

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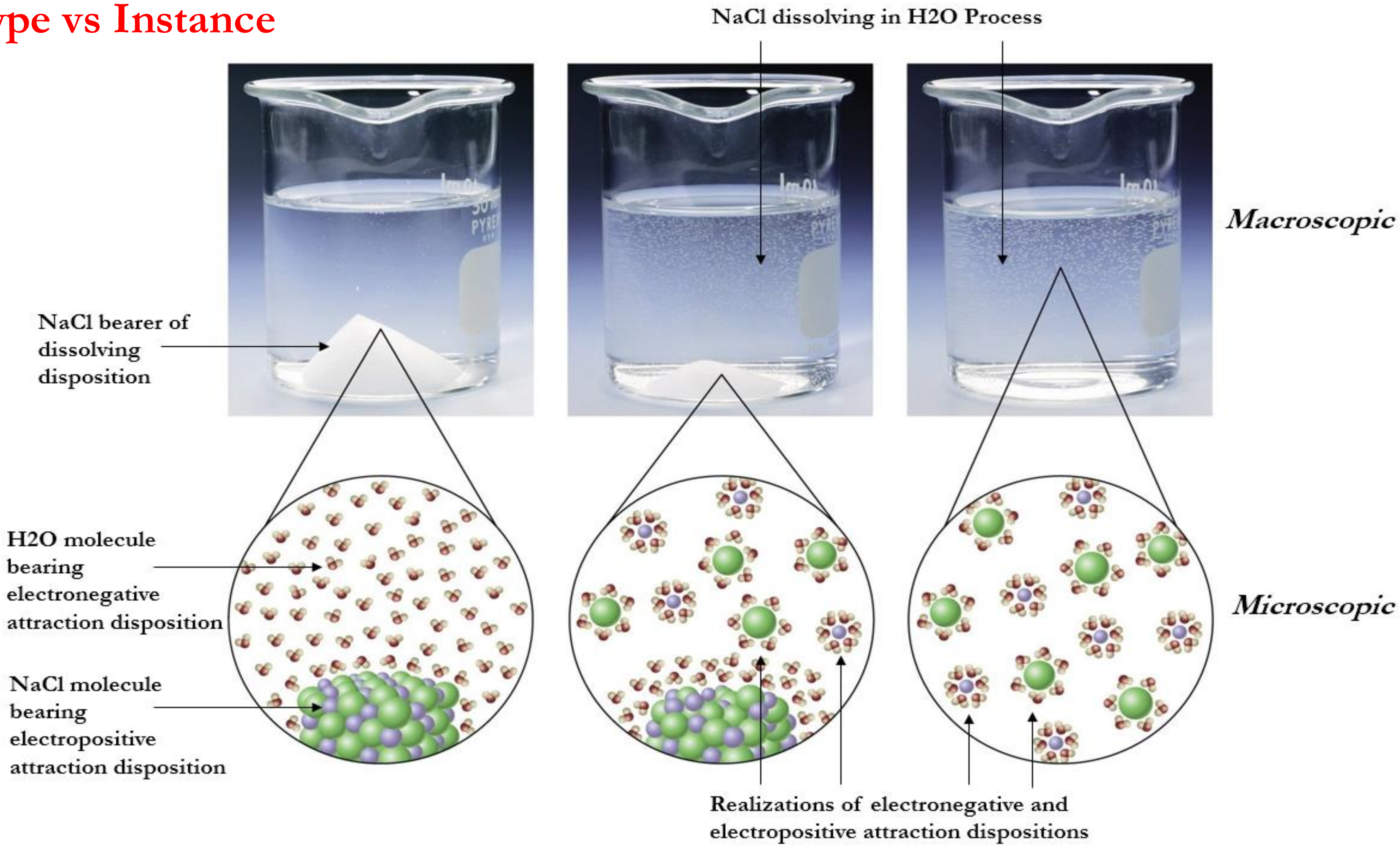
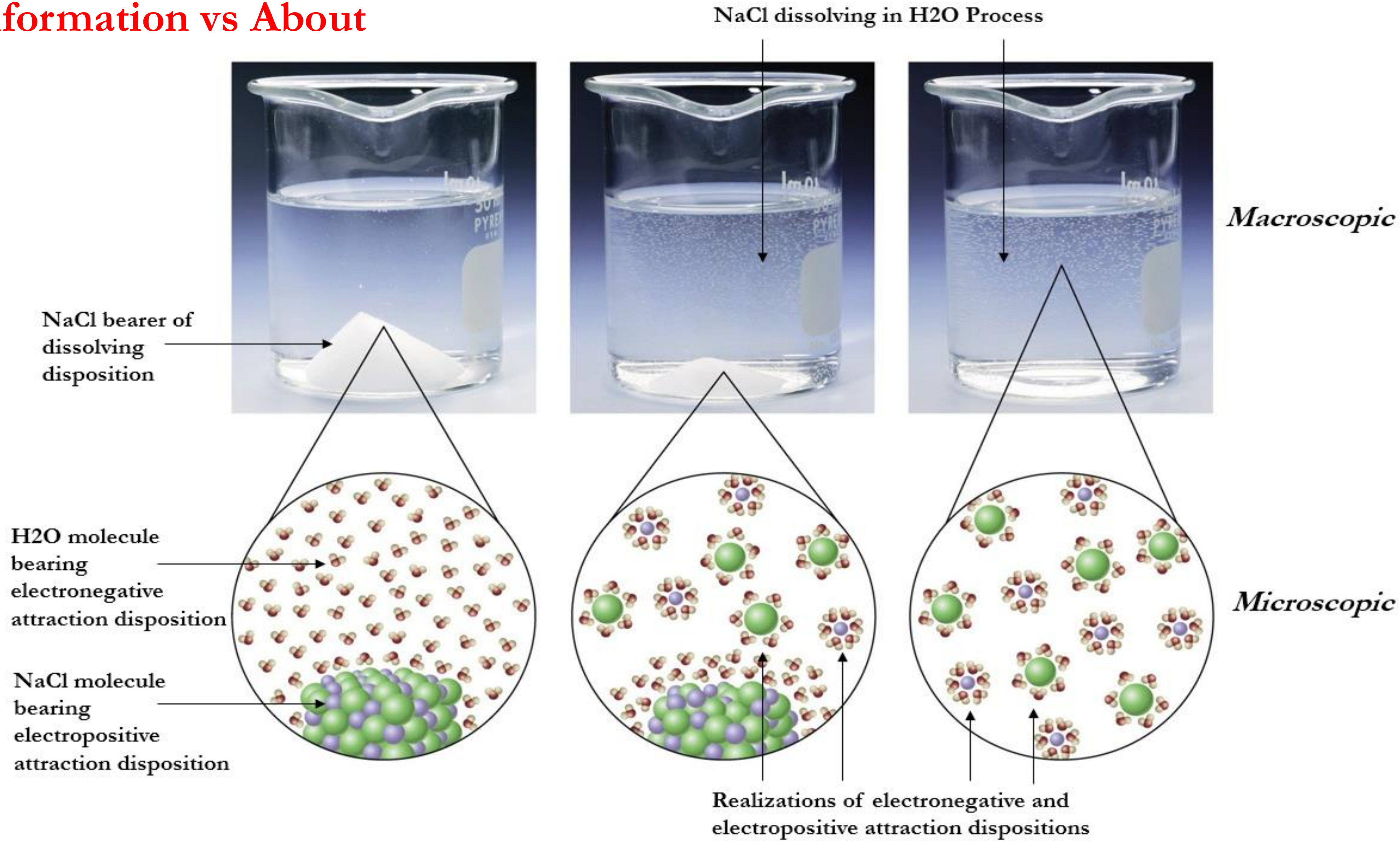
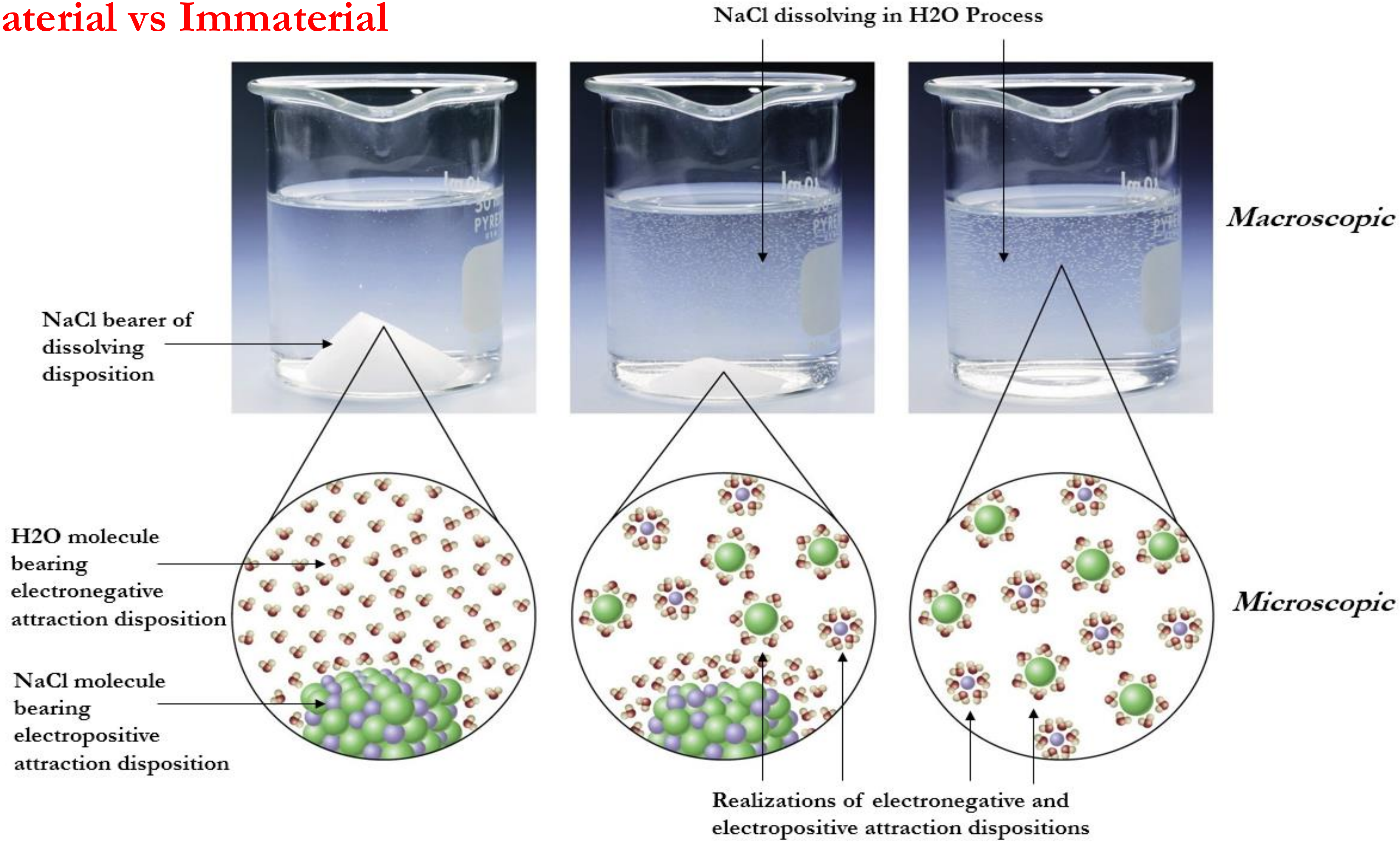
Type vs Instance



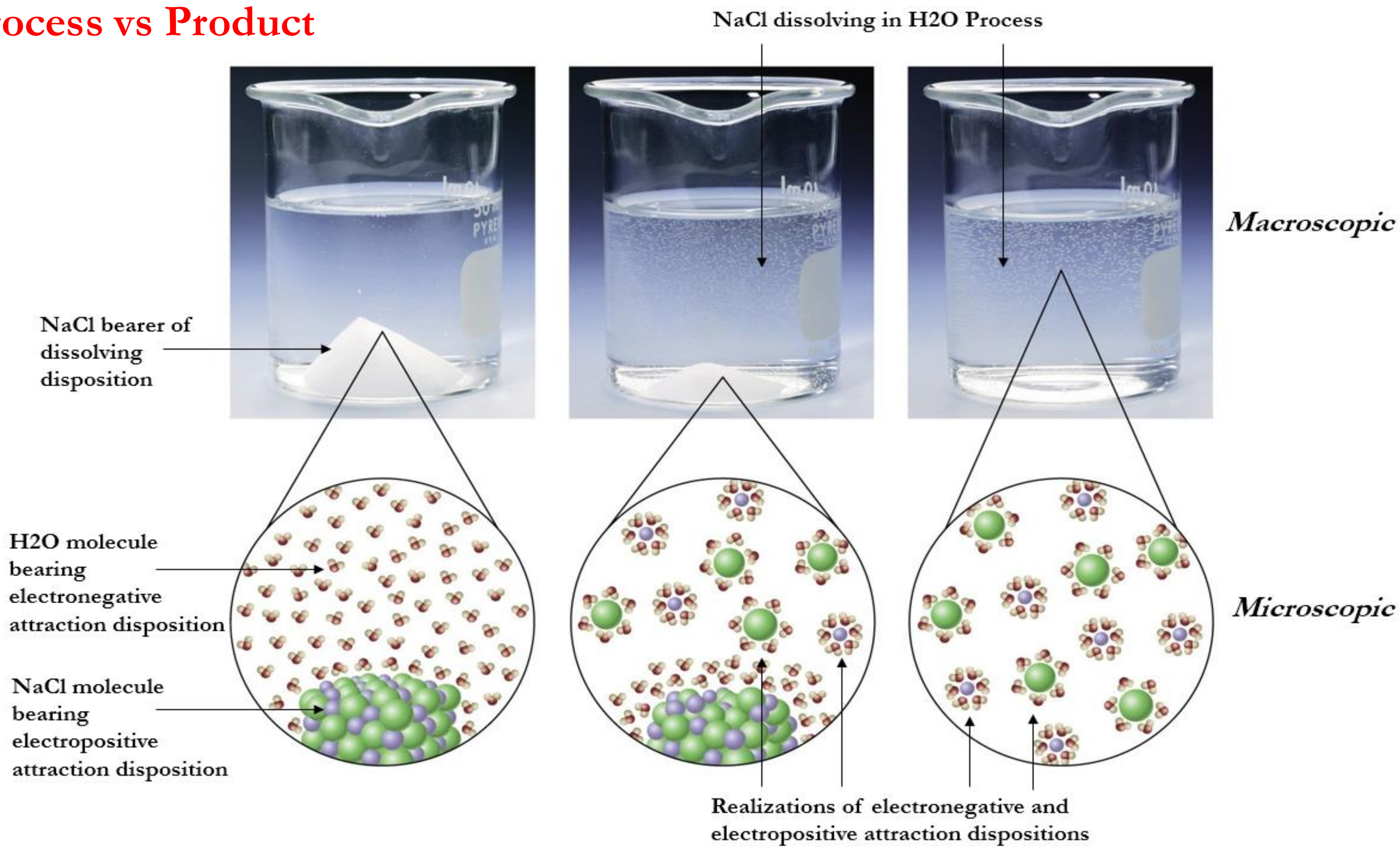
Information vs About



Material vs Immaterial



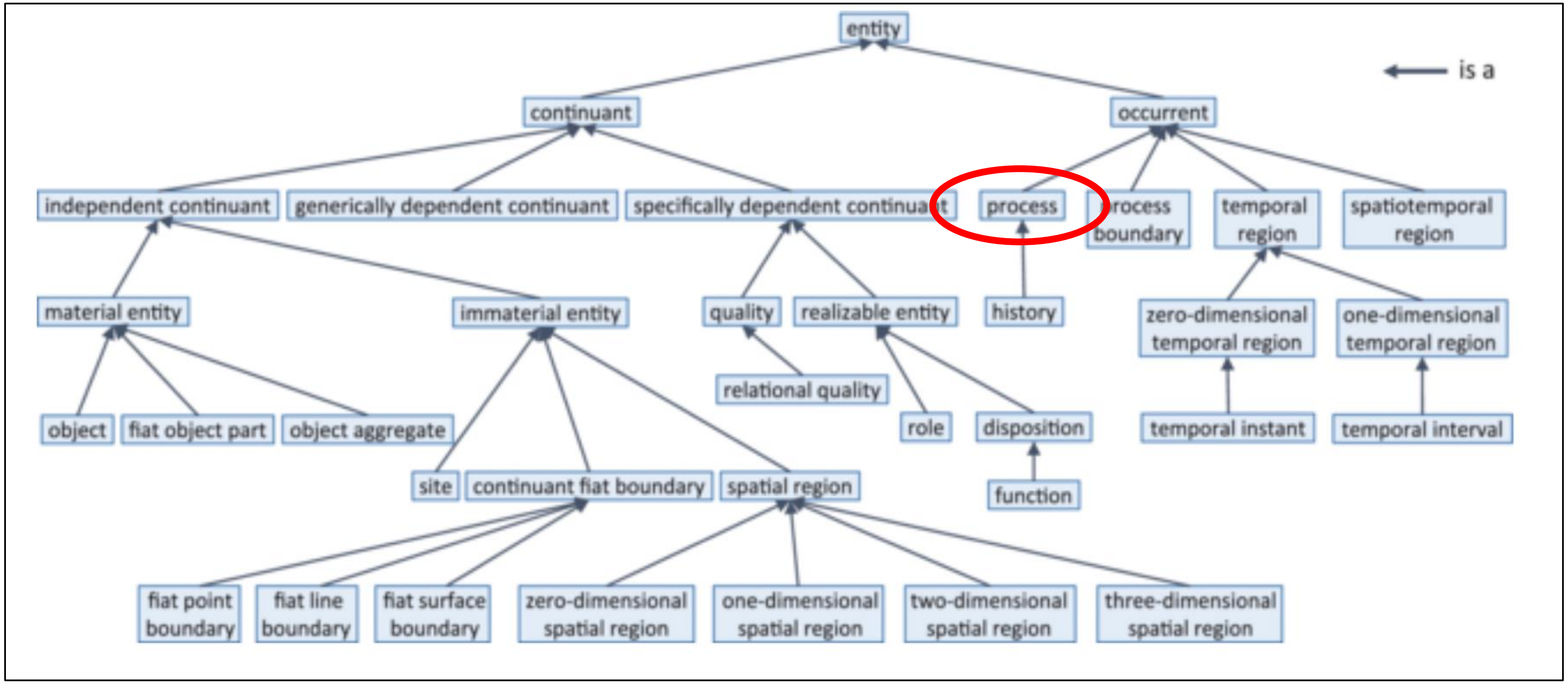
Process vs Product



BECAUSE THESE POINTS ARE SO OBVIOUS, THEY ARE SO OFTEN OVERLOOKED

Dependency Chain

- A given **process** may have realization some realizable entity, which inheres in some material entity



Dependency Chain

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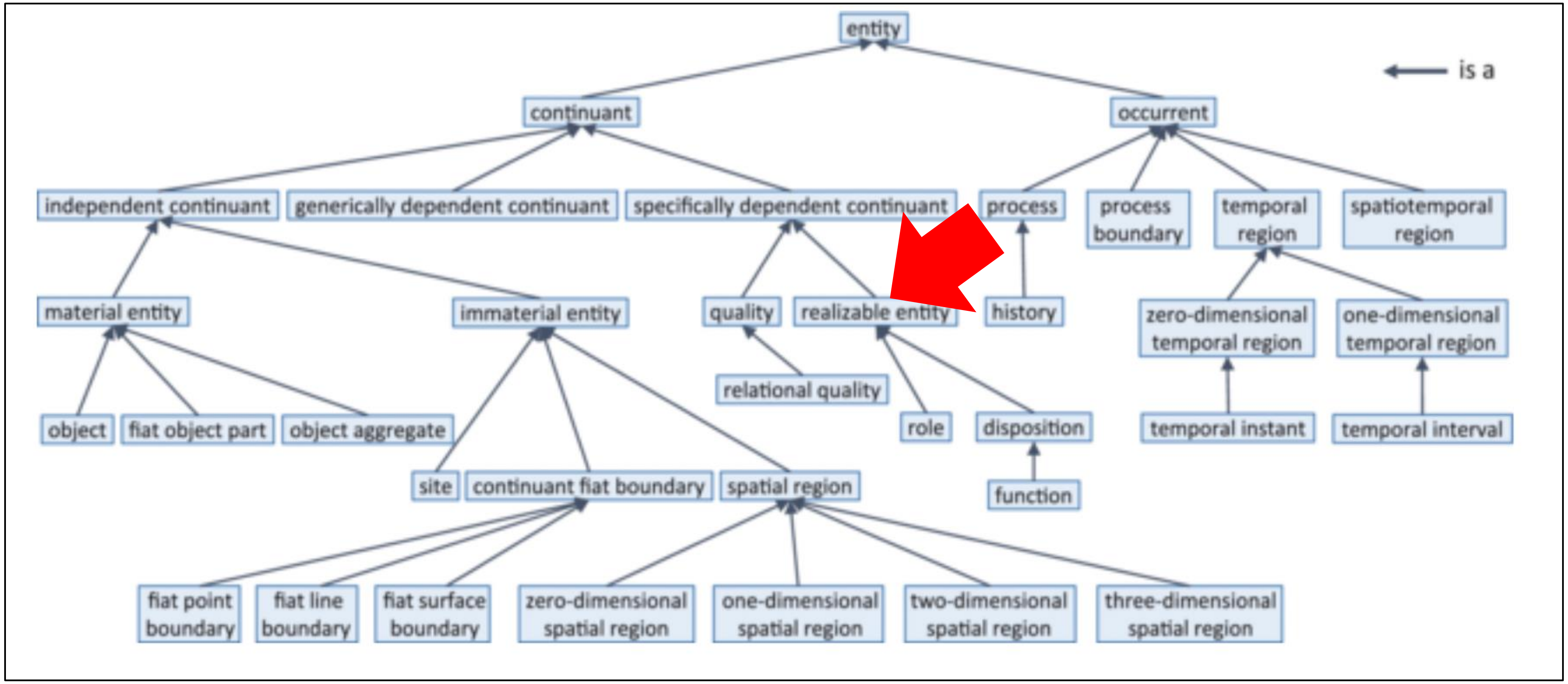

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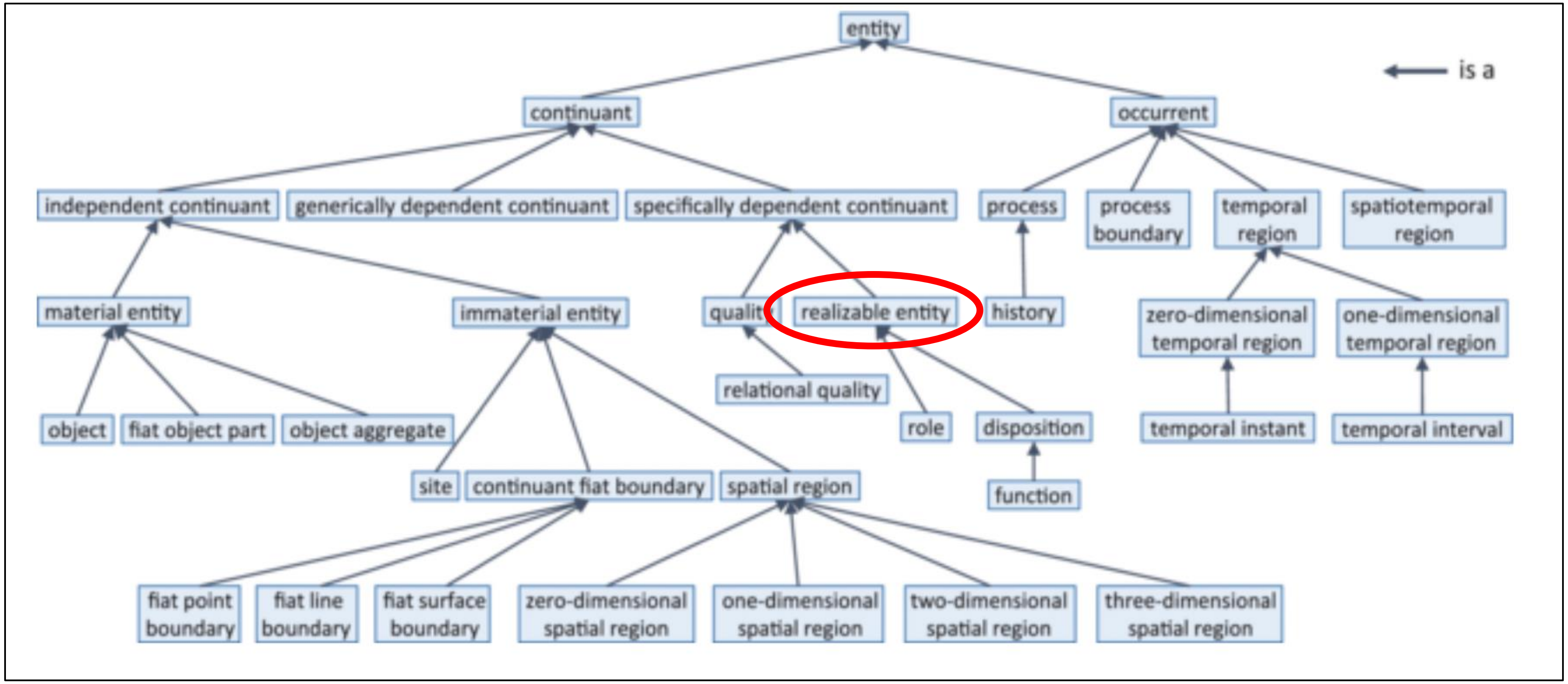

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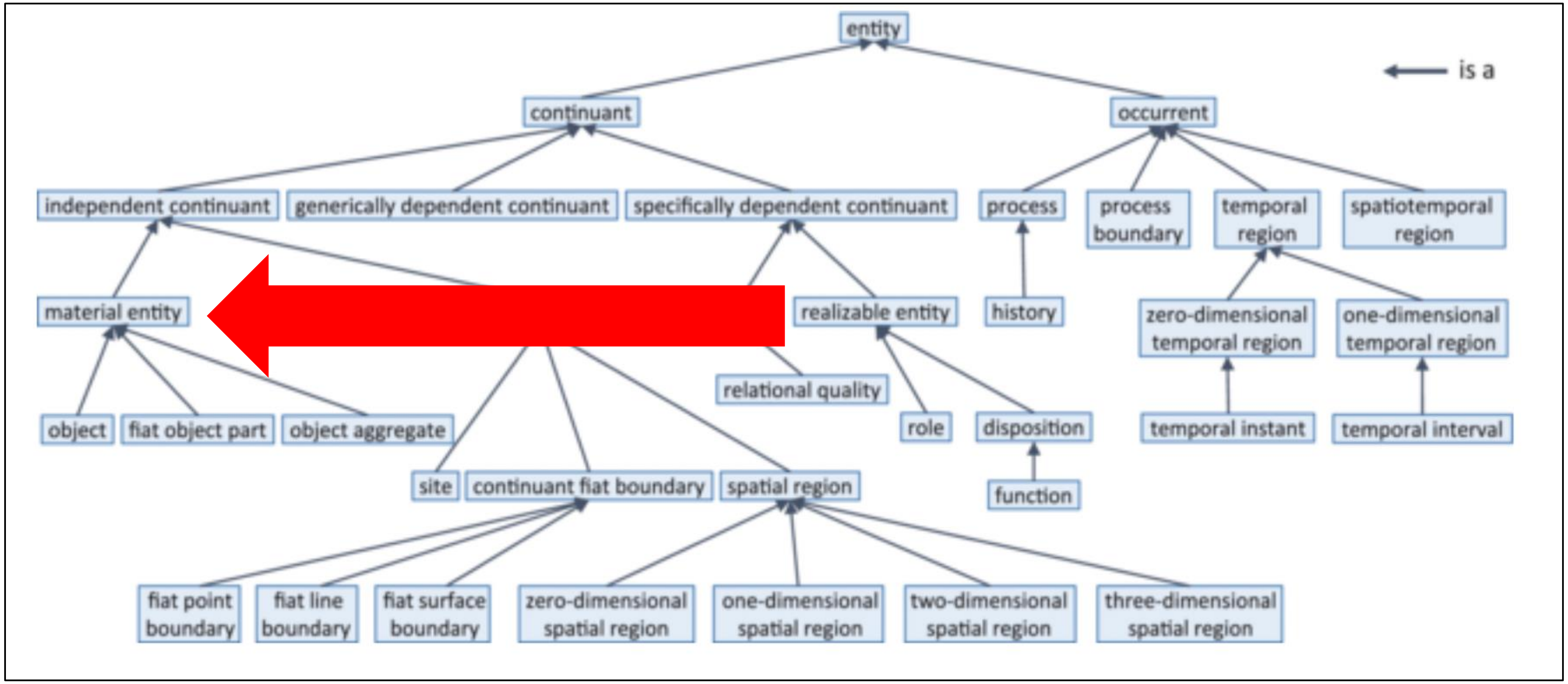

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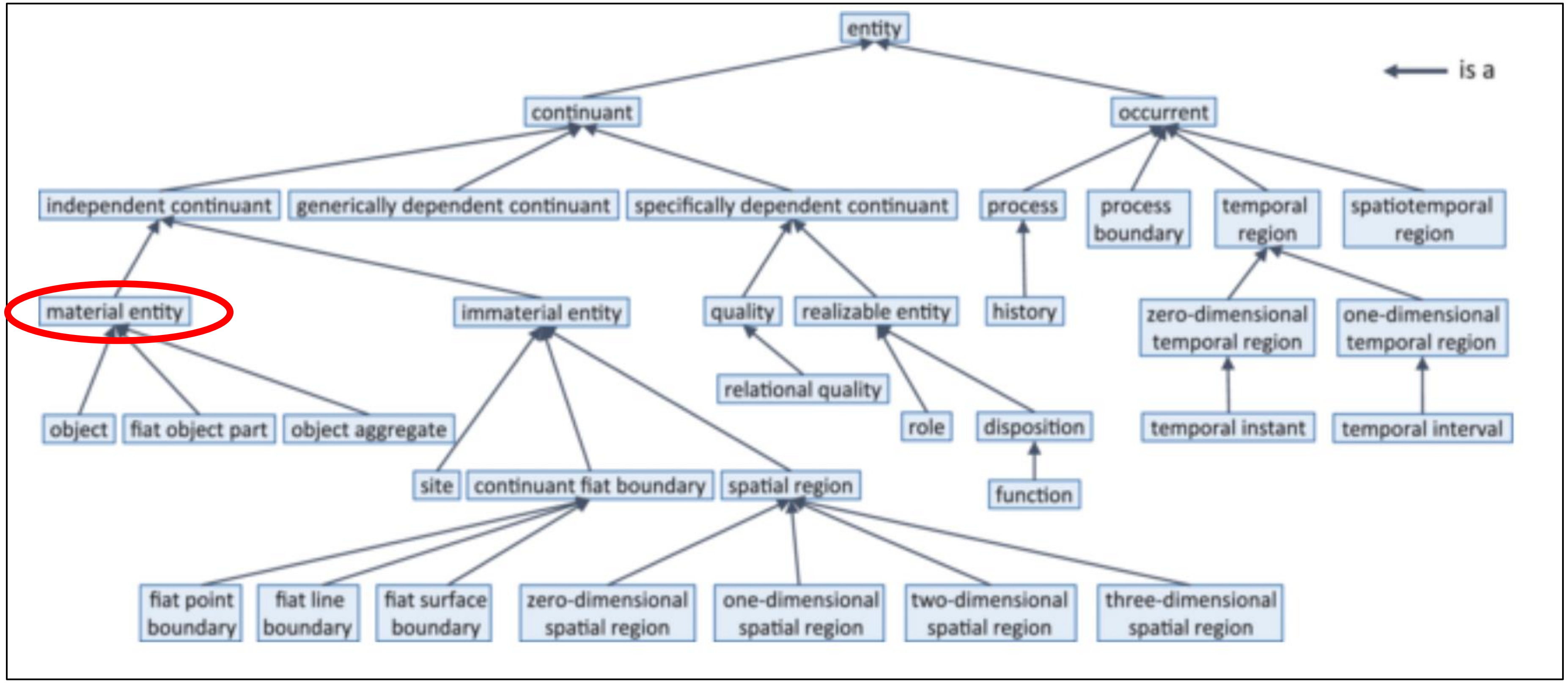
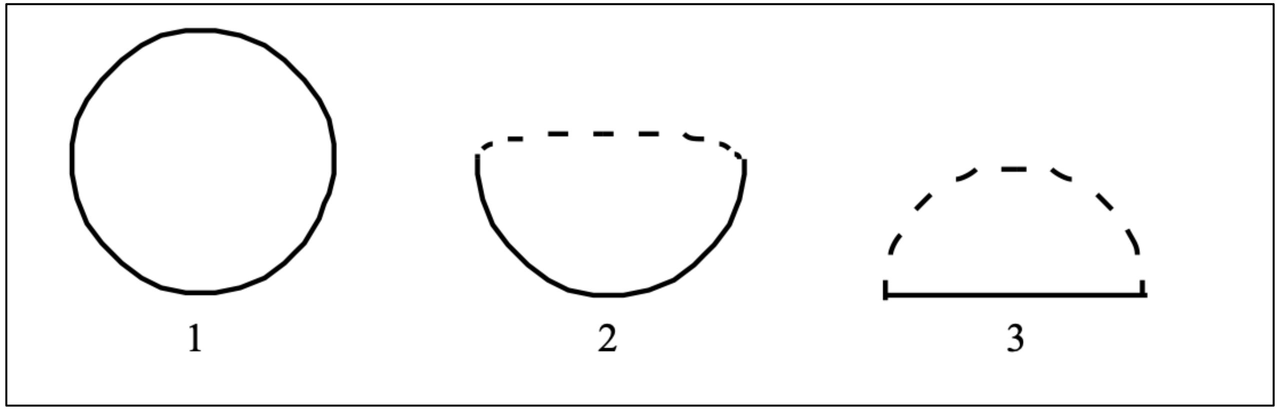


Site

- Immaterial entities whose locations are determined in relation to some material entity

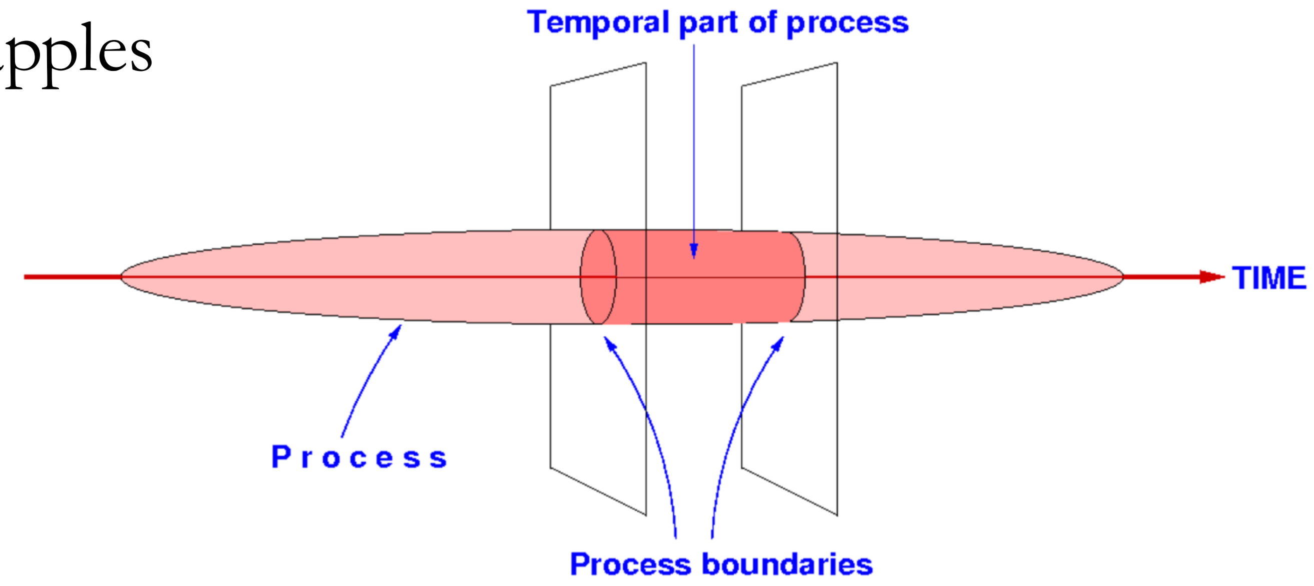
- Examples:
  - A rabbit hole
  - The interior of your bedroom
  - The hold of a ship
  - The cockpit of an aircraft



Temporal Region

- The process of baking of an apple pie has proper parts, such as cutting apples, preparing pastry crust, etc.

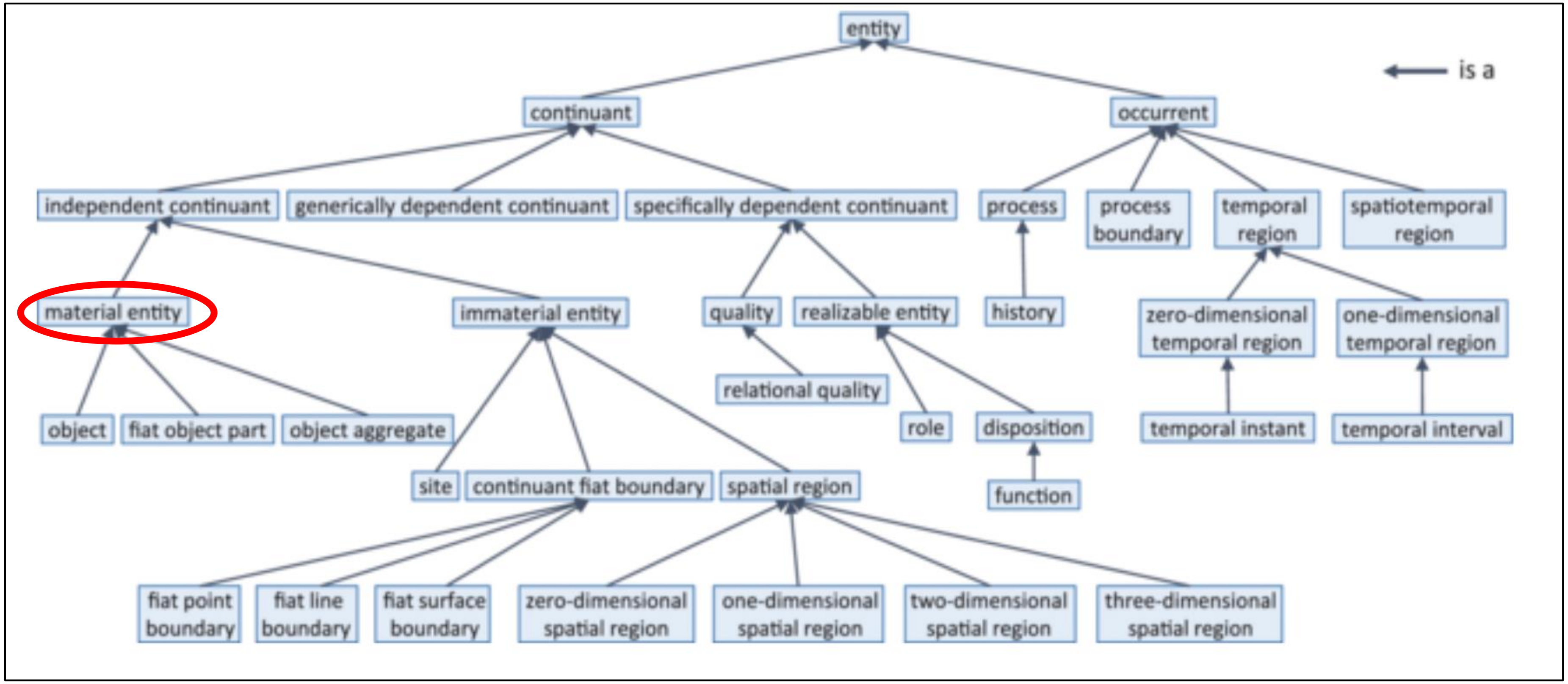
- Proper process parts may be further divided, e.g. cutting of specific apples



* *Image from Galton, 2016: Processes and Events in BFO and DOLCE*

Dependency Chain

- A given **material entity** is located at some site at some time all of which is described by some information



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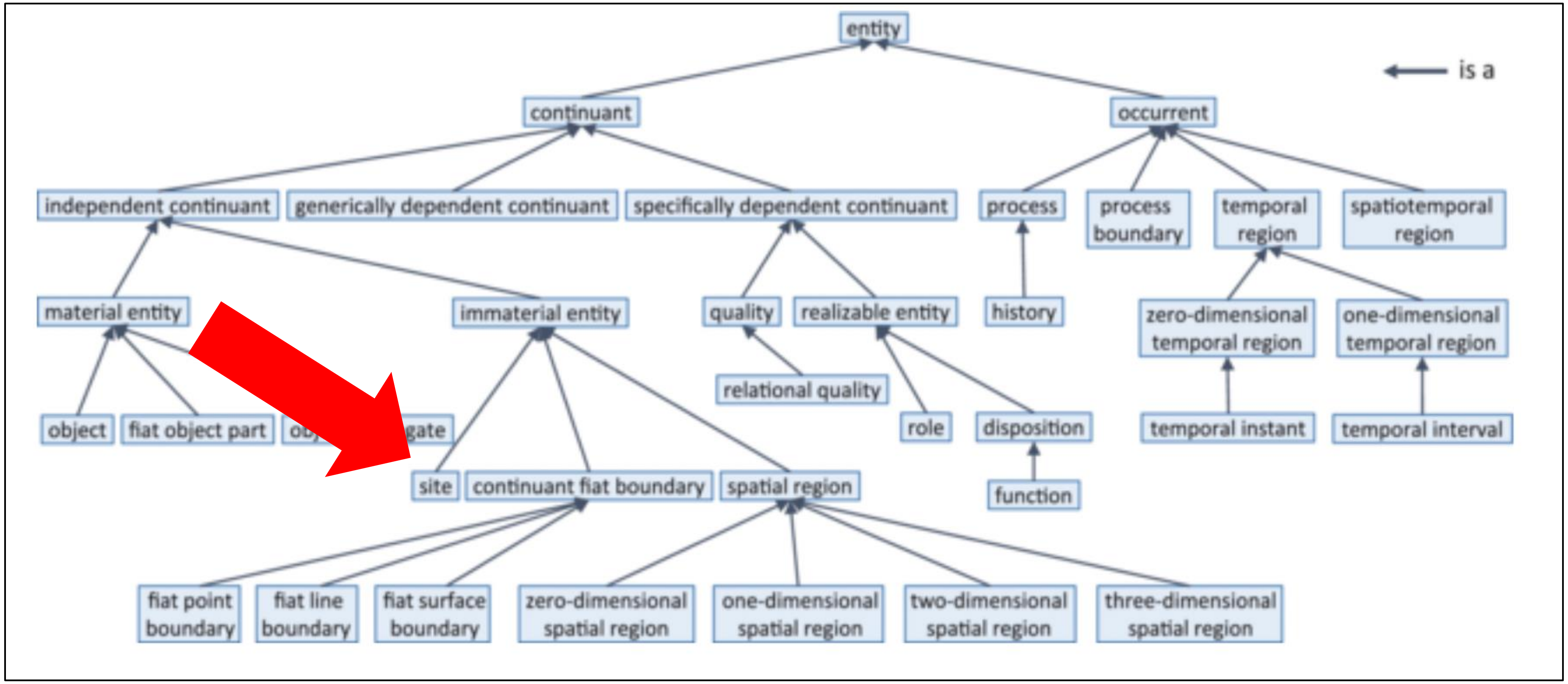

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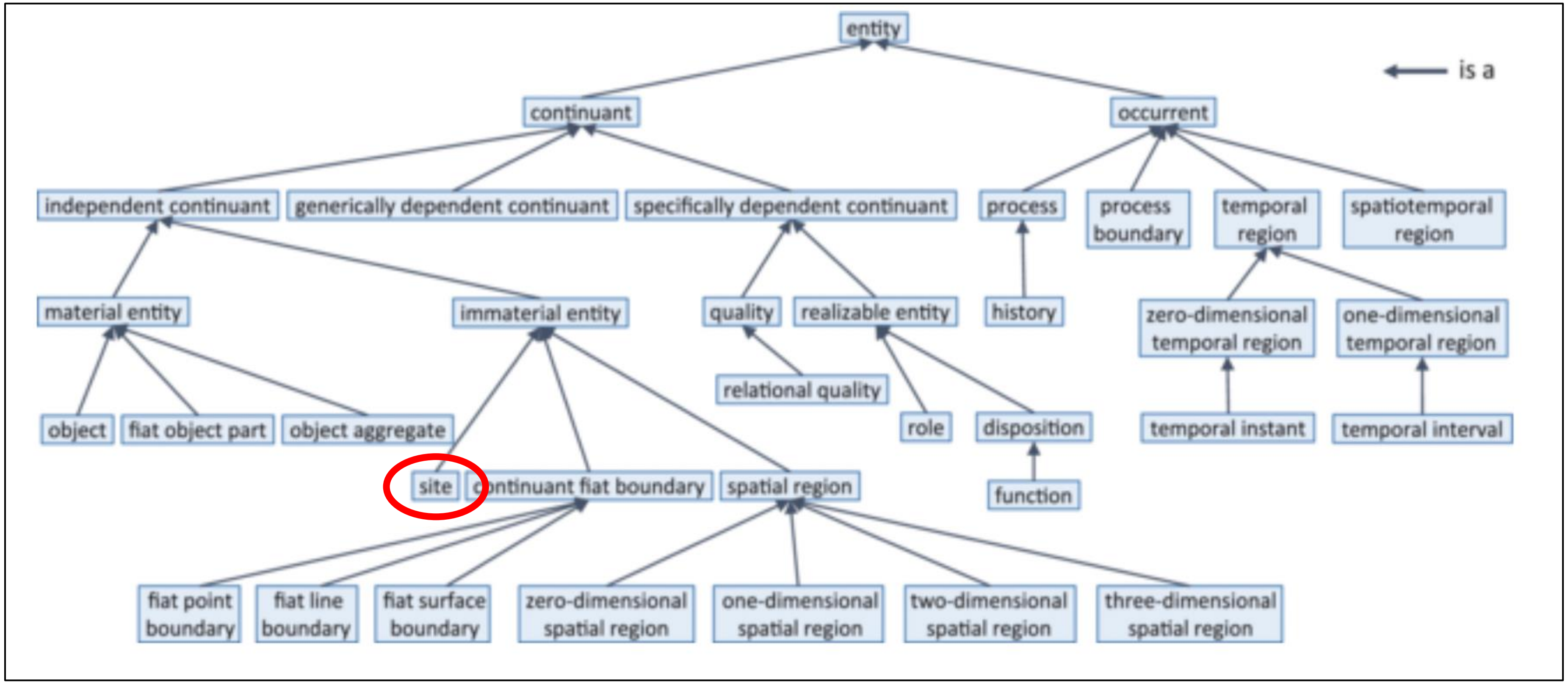

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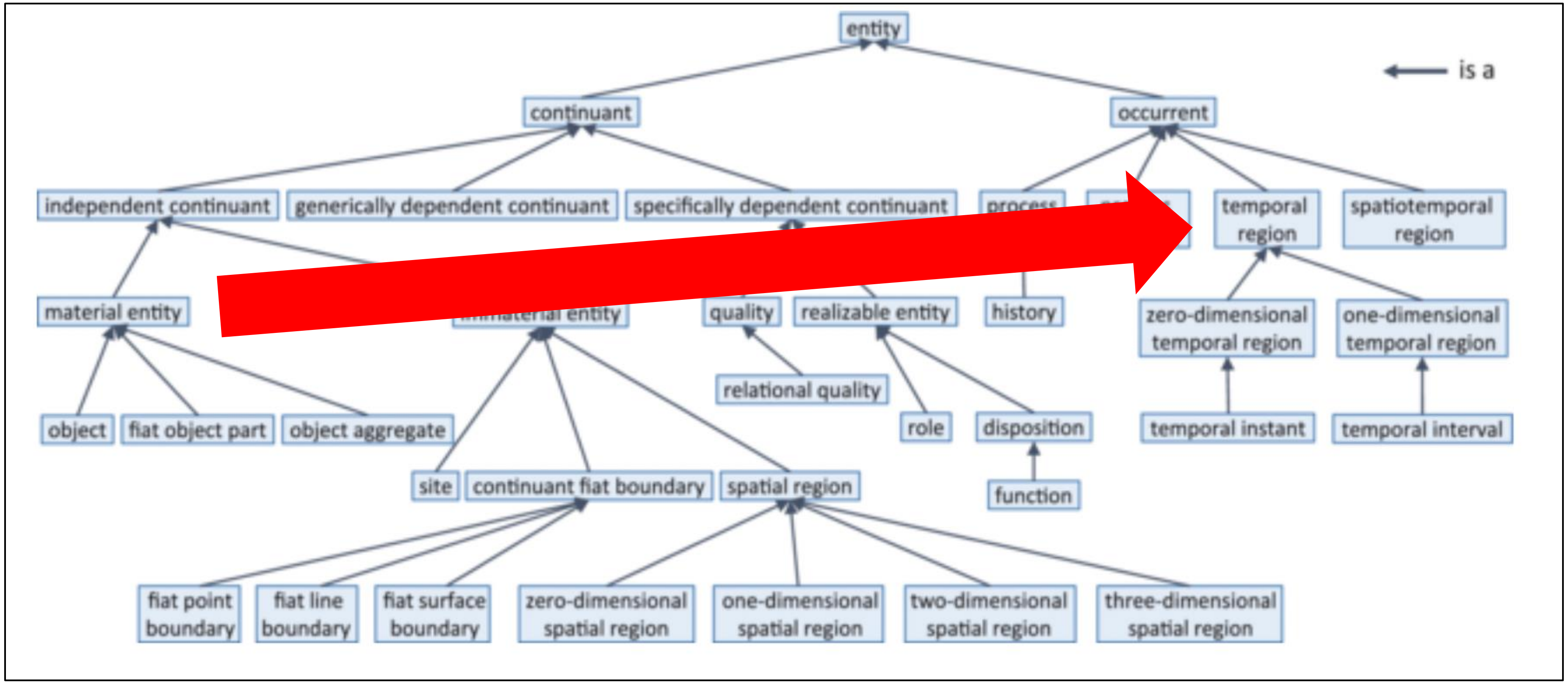

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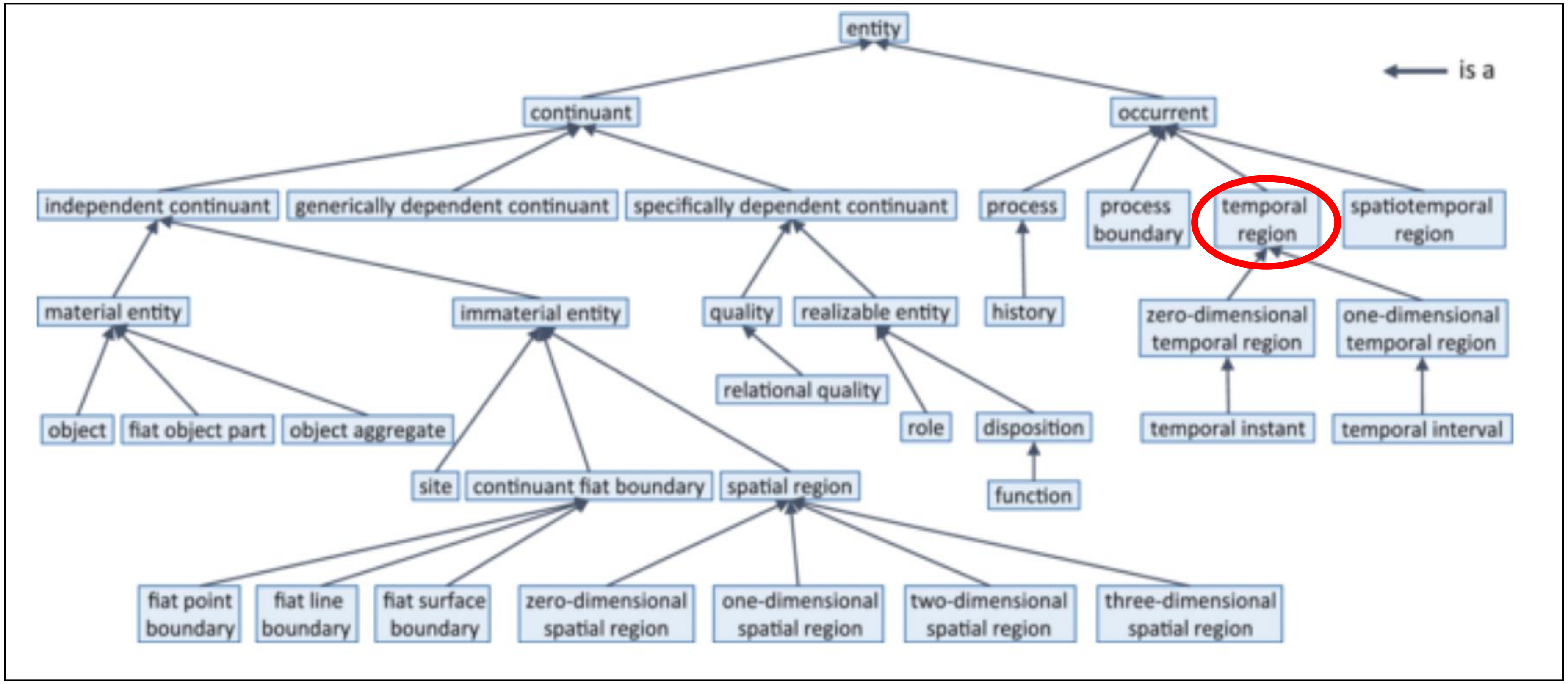


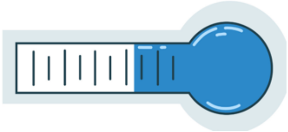
Information

Material Entity

apple

Information

temp measure



Many measures are about material entities

Dependency Chain

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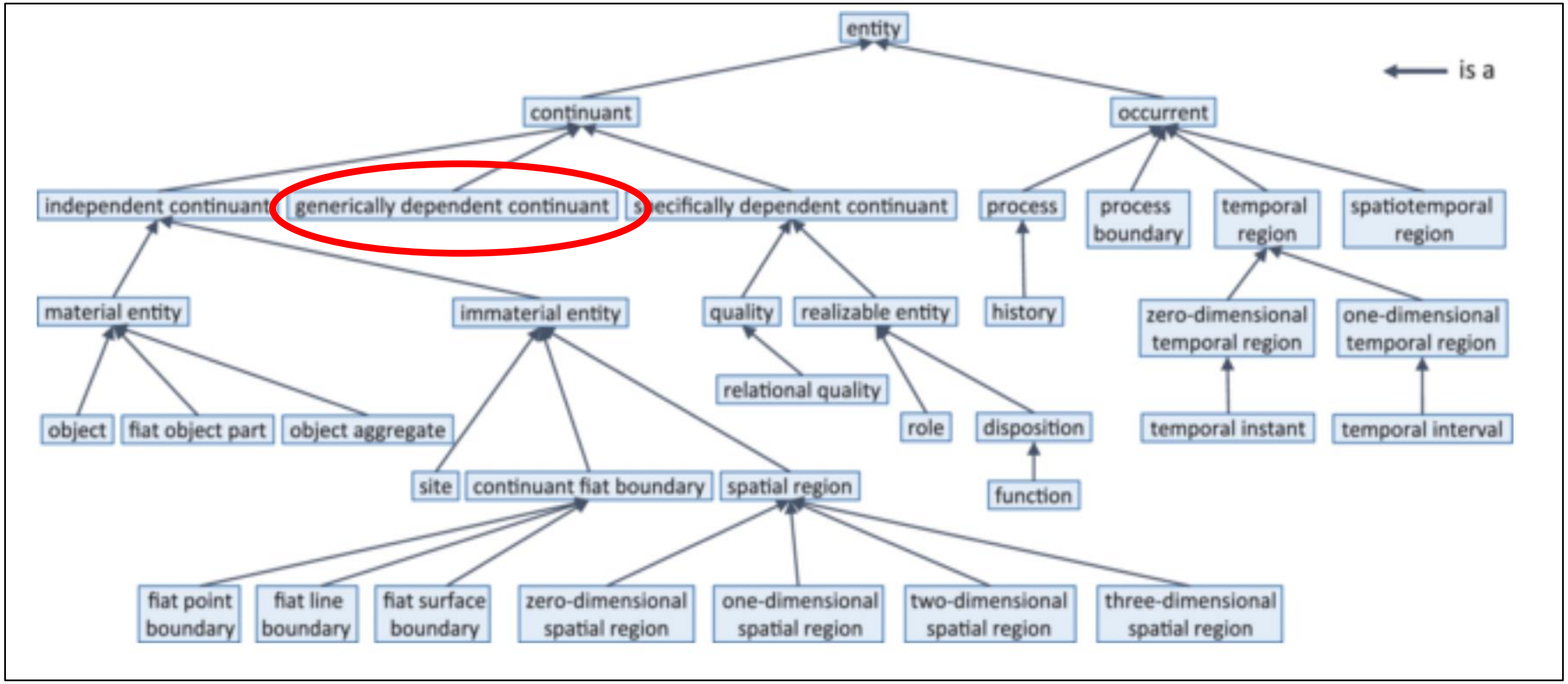

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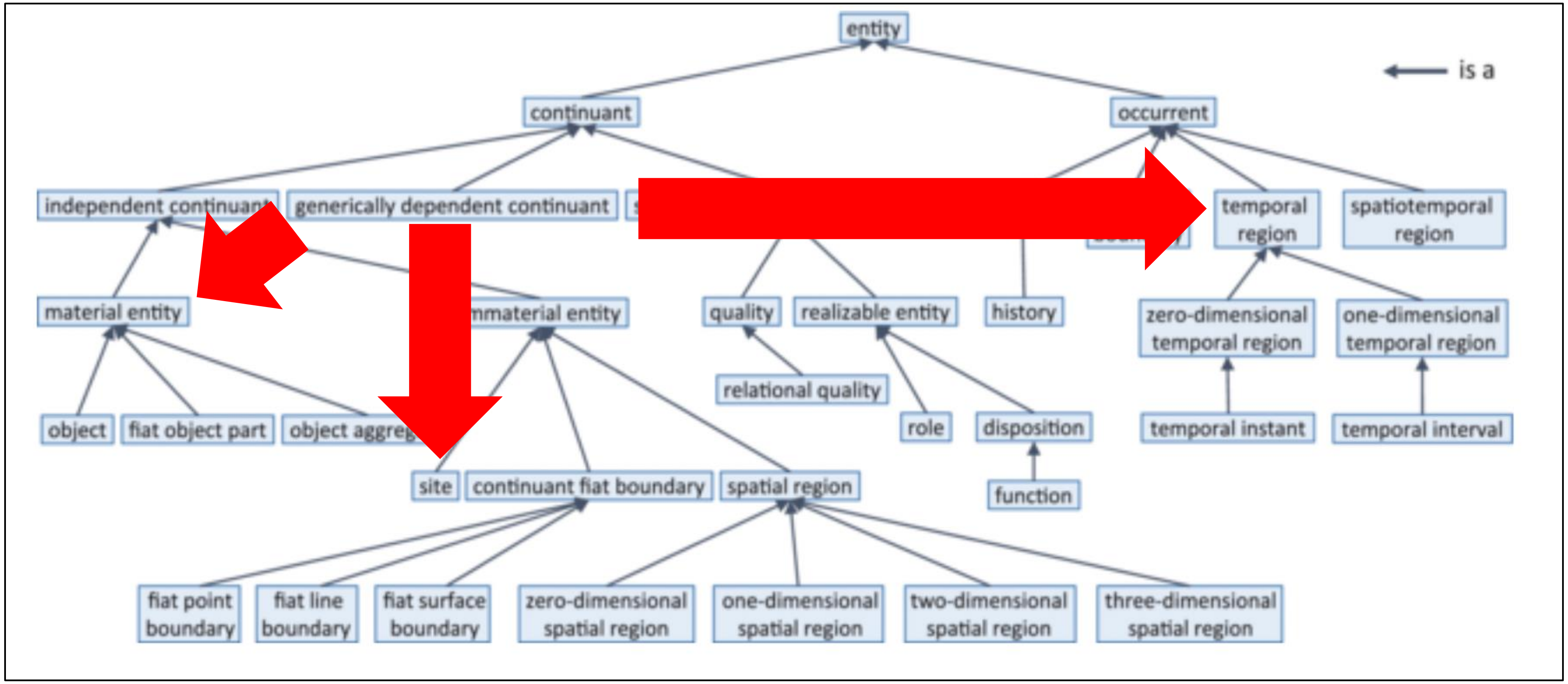


Guidance

- Competency questions are **used to guide ontology development** and **generate unit tests** to ensure ontologies are sufficiently well-developed

- Identify a preliminary list of competency questions **first**

- Sometimes called "decision support questions" or "requirements" or etc.

Competency Questions

Classes & Relations

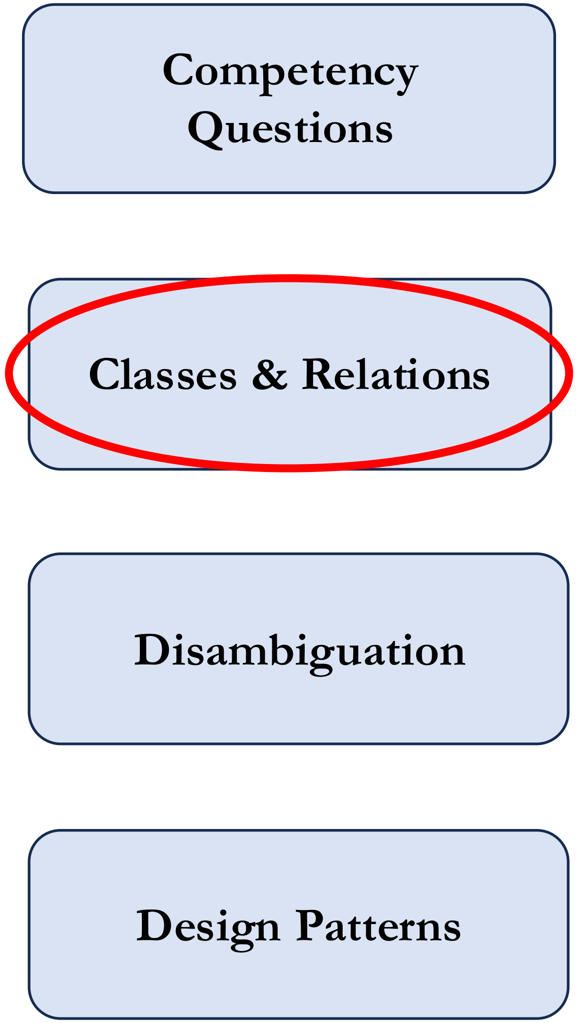
Disambiguation

Design Patterns

At what speed does a patrol boat move in knots over an hour?

Classes & Relations

- Using competency questions as guidance, list the classes and relations you will need to represent to answer the questions
- Create this list through the lens of, say, Basic Formal Ontology (BFO) and/or Common Core Ontologies classes and relations
- I illustrate using BFO



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    - generically dependent continuant
    - independent continuant
      - immaterial entity
        - continuant fiat boundary
          - fiat line
          - fiat point
          - fiat surface
        - site
        - spatial region
          - one-dimensional spatial region
          - three-dimensional spatial region
          - two-dimensional spatial region
          - zero-dimensional spatial region
      - material entity
        - fiat object part
        - object
        - object aggregate
    - specifically dependent continuant
      - quality
        - relational quality
      - realizable entity
        - disposition
          - function
        - role
  - occurrent
    - process
      - history
    - process boundary
    - spatiotemporal region
    - temporal region
      - one-dimensional temporal region
        - temporal interval
      - zero-dimensional temporal region
        - temporal instant

- concretizes
- continuant part of
  - member part of
- environs
- exists at
- first instant of
- generically depends on
- has continuant part
  - has member part
- has first instant
- has history
- has last instant
- has material basis
- has occurrent part
  - has temporal part
- has participant
- has realization
- history of
- is carrier of
- is concretized by
- last instant of
- located in
- location of
- material basis of
- occupies spatial region
- occupies spatiotemporal region
- occupies temporal region
- occurrent part of
  - temporal part of
- occurs in
- participates in
- preceded by
- precedes
- realizes
- spatially projects onto
- specifically depended on by
  - bearer of
- specifically depends on
  - inheres in
- temporally projects onto

Operationalizing BFO

- When identifying entities, describe:

1. What has material parts, i.e. **Material Entity**
2. What qualities these material entities have, i.e. **Quality**
3. What these material entities might do, i.e. **Realizable Entity**
4. What these material entities actually do, i.e. **Process**
5. Where these material entities are located, i.e. **Sites**
6. When these entities exist, i.e. **Temporal Region**
7. Information we use to talk about 1-6, i.e. **Information**

https://github.com/BFO-ontology/BFO-2020

Classes

- Material Entities –
- Qualities –
- Processes –
- Realizables –
- Sites –
- Temporal Region –
- Information –

At what speed does a patrol boat move in knots over an hour?

Classes

- Material Entities – **Patrol boat**
- Qualities –
- Processes –
- Realizables –
- Sites –
- Temporal Region –
- Information –

At what speed does a patrol boat move in knots over an hour?

Classes

- Material Entities – Patrol boat
- Qualities –
- Processes – **Act of motion**
- Realizables –
- Sites –
- Temporal Region –
- Information –

At what speed does a patrol boat move in knots over an hour?

Classes

- Material Entities – Patrol boat
- Qualities –
- Processes – Act of motion, **speed?**
- Realizables –
- Sites –
- Temporal Region –
- Information – **speed?**

At what speed does a patrol boat move in knots over an hour?

Classes

- Material Entities – Patrol boat
- Qualities –
- Processes – Act of motion, speed*
- Realizables –
- Sites –
- Temporal Region –
- Information – speed*

use * to note ambiguity then move on; we will revisit

At what speed does a patrol boat move in knots over an hour?

Classes

- Material Entities – Patrol boat
- Qualities –
- Processes – Act of motion, speed*
- Realizables –
- Sites –
- Temporal Region –
- Information – speed*, **knots measurement**

At what speed does a patrol boat move in knots over an hour?

Classes

- Material Entities – Patrol boat
- Qualities –
- Processes – Act of motion, speed*
- Realizables –
- Sites –
- Temporal Region – **hours***
- Information – speed*, knots measurement, **hours***

use * to note ambiguity then move on; we will revisit

At what speed does a patrol boat move in knots over an hour?

Disambiguate

- Logic is **demanding**, in part because it is **complete**
- We make explicit the implicit semantics within data, which requires disambiguating
- It is easier to stitch meaning together having cut it from whole cloth, than it is from disparate meanings

Competency Questions

Classes & Relations

Disambiguation

Design Patterns

Disambiguation

- **Information** vs what that information **is about**, e.g. occupation code vs a holder of an occupation

- **Material** vs **immaterial** things, e.g. a given river vs the site where the river used to flow

- **Bearing properties** vs **bearers of properties**, e.g. apple's redness vs the apple

- **Processes** vs **product**, e.g. ontology engineering vs ontology produced

Revisiting Ambiguity

- “speed” as a process vs information about a process

Revisiting Ambiguity

- “speed” as a process vs information about a process

- Speed is the magnitude of a change in position over time

INFORMATION

Revisiting Ambiguity

- “speed” as a process vs information about a process

- Speed is the changing of position over time

PROCESS

Revisiting Ambiguity

- "speed" as a process vs information about a process

At what speed does a patrol boat move in knots over an hour?

WHICH DO WE CARE ABOUT FOR THIS COMPETENCY QUESTION?

Simplify

- Material Entities – Patrol boat
- Qualities –
- Processes – Act of motion, speed*
- Realizables –
- Sites –
- Temporal Region – hours*
- Information – speed*, knots measurement, hours*

At what speed does a patrol boat move in knots over an hour?

Simplify

- Material Entities – Patrol boat
- ~~Qualities –~~
- Processes – Act of motion, speed*
- ~~Realizables –~~
- ~~Sites –~~
- Temporal Region – hours*
- Information – ~~speed*,~~ knots measurement, ~~hours*~~

simplify the list

At what speed does a patrol boat move in knots over an hour?

Relations

- Material Entities – Patrol boat
- Processes – Act of motion, speed
- Temporal Region – hours
- Information – knots measurement

and reflect on relationships among the listed entities

At what speed does a patrol boat move in knots over an hour?

Operationalizing BFO

- When identifying relations, describe:

1. Qualities to material entities, i.e. **inheres in**
2. Realizables to material entities, i.e. **inheres in**
3. Processes to material entities, i.e **participates in**
4. Realizables to processes, i.e. **has realization**
5. Immaterial location of material entity, i.e. **located in**
6. When any such entities exist, i.e. **exists at**
7. How any such entities relate to information, e.g. **is about**

https://github.com/BFO-ontology/BFO-2020

Operationalizing BFO

- When identifying relations, describe:

    1. ~~Qualities to material entities, i.e. **inheres in**~~
    2. ~~Realizables to material entities, i.e. **inheres in**~~
    3. Processes to material entities, i.e **participates in**
    4. ~~Realizables to processes, i.e. **has realization**~~
    5. ~~Immaterial location of material entity, i.e. **located in**~~
    6. When any such entities exist, i.e. **exists at**
    7. How any such entities relate to information, e.g. **is about**

https://github.com/BFO-ontology/BFO-2020

Simplify

- When identifying relations, describe:

  1. Processes to material entities, i.e **participates in**
  2. How any such entities relate to information, e.g. **is about**
  3. When any such entities exist, i.e. **exists at**

[*https://github.com/BFO-ontology/BFO-2020*](https://github.com/BFO-ontology/BFO-2020)

Relations

- Material Entities – **Patrol boat**
- Processes – **Act of motion**, speed
- Temporal Region – hours
- Information – knots measurement

patrol boats participate in processes

- concretizes
- continuant part of
  - member part of
- environs
- exists at
- first instant of
- generically depends on
- has continuant part
  - has member part
- has first instant
- has history
- has last instant
- has material basis
- has occurrent part
  - has temporal part
- has participant
- has realization
- history of
- is carrier of
- is concretized by
- last instant of
- located in
- location of
- material basis of
- occupies spatial region
- occupies spatiotemporal region
- occupies temporal region
- occurrent part of
  - temporal part of
- occurs in
- participates in
- preceded by
- precedes
- realizes
- spatially projects onto
- specifically depended on by
  - bearer of
- specifically depends on
  - inheres in
- temporally projects onto

[*https://github.com/BFO-ontology/BFO-*](https://github.com/BFO-ontology/BFO-)

Recipe

- Material Entities – Patrol boat
- Processes – **Act of motion**, speed
- Temporal Region – **hours**
- Information – knots measurement

motion exists at some temporal region

Recipe

- Material Entities – Patrol boat
- Processes – Act of motion, **speed**
- Temporal Region – hours
- Information – **knots measurement**

knots measurement is about speed

Design Patterns

- Classes and relations identified, turn next to constructing visual representations reflecting the competency questions

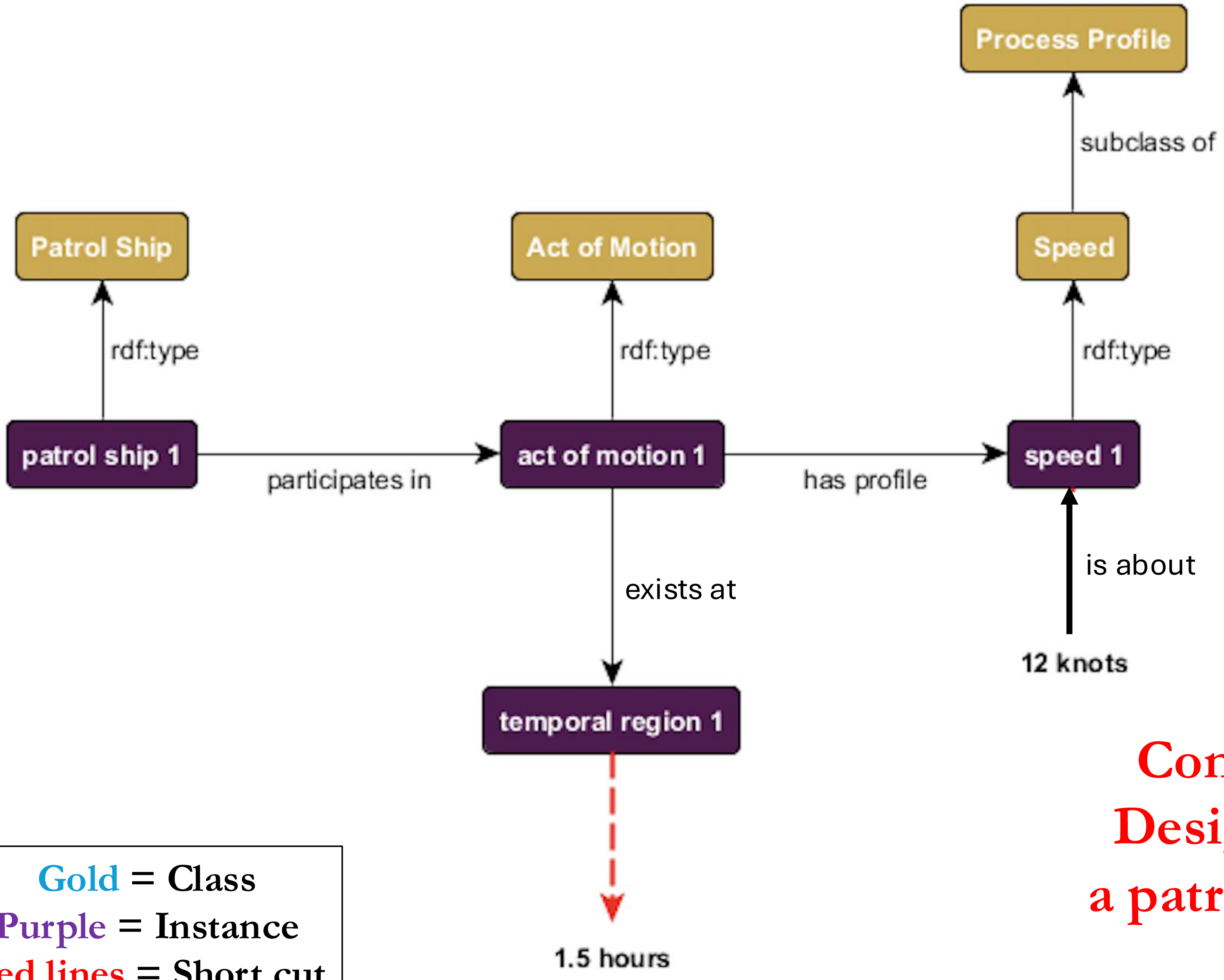
- If you have completed the preceding steps, this should be relatively straightforward

Competency Questions

Classes & Relations

Disambiguation

Design Patterns



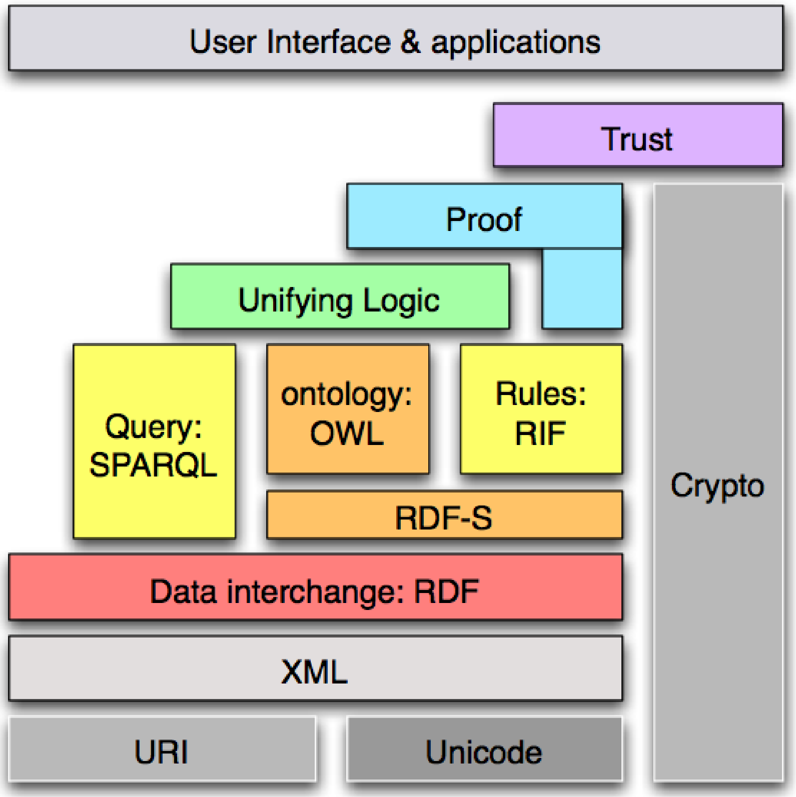
Gold = Class
Purple = Instance
Red lines = Short cut

Common Core Ontologies Design Pattern representing a patrol ship traveling 12 knots over 1.5 hours

Outline

- How Field are Born
- New Era, New Tradecraft
- Tradecraft as Recipe
- Tradecraft as Encoding

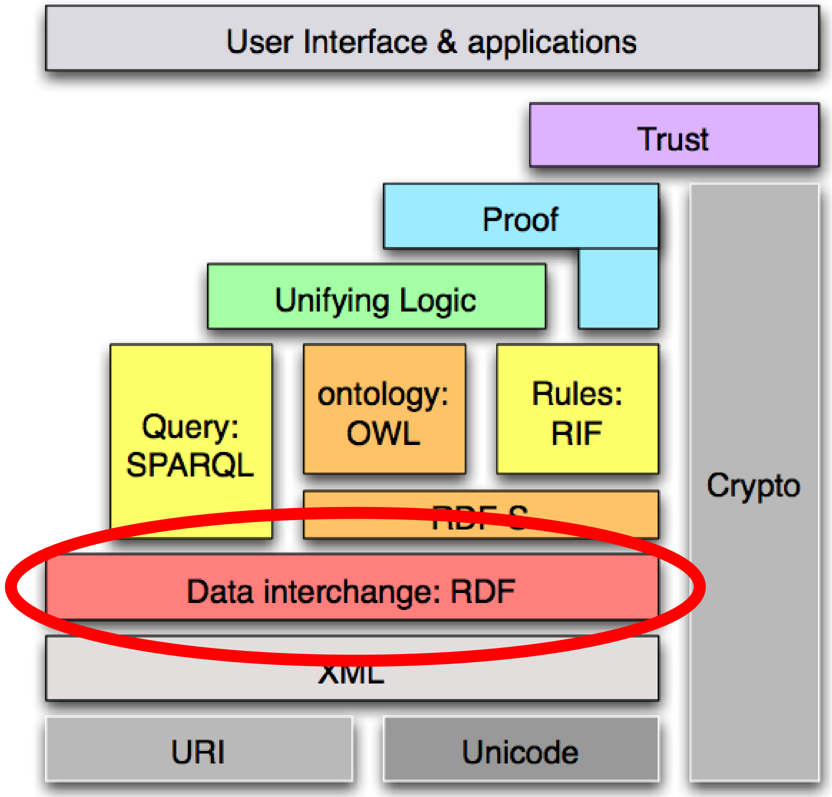
Semantic Web Stack



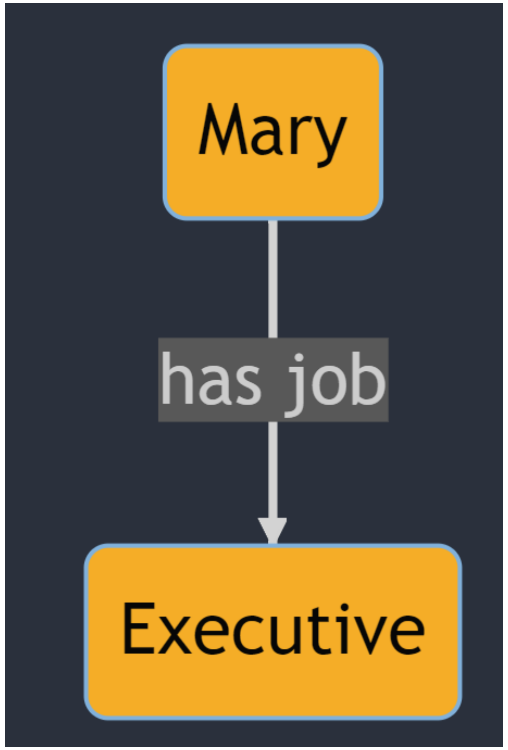
Semantic Web Stack

- "RDF" stands for:
  - **Resource**: Everything that can have a unique identifier, e.g. pages, places, people, dogs, products...
  - **Description**: attributes, features, and relations of among resources
  - **Framework**: model, languages and syntaxes for these descriptions

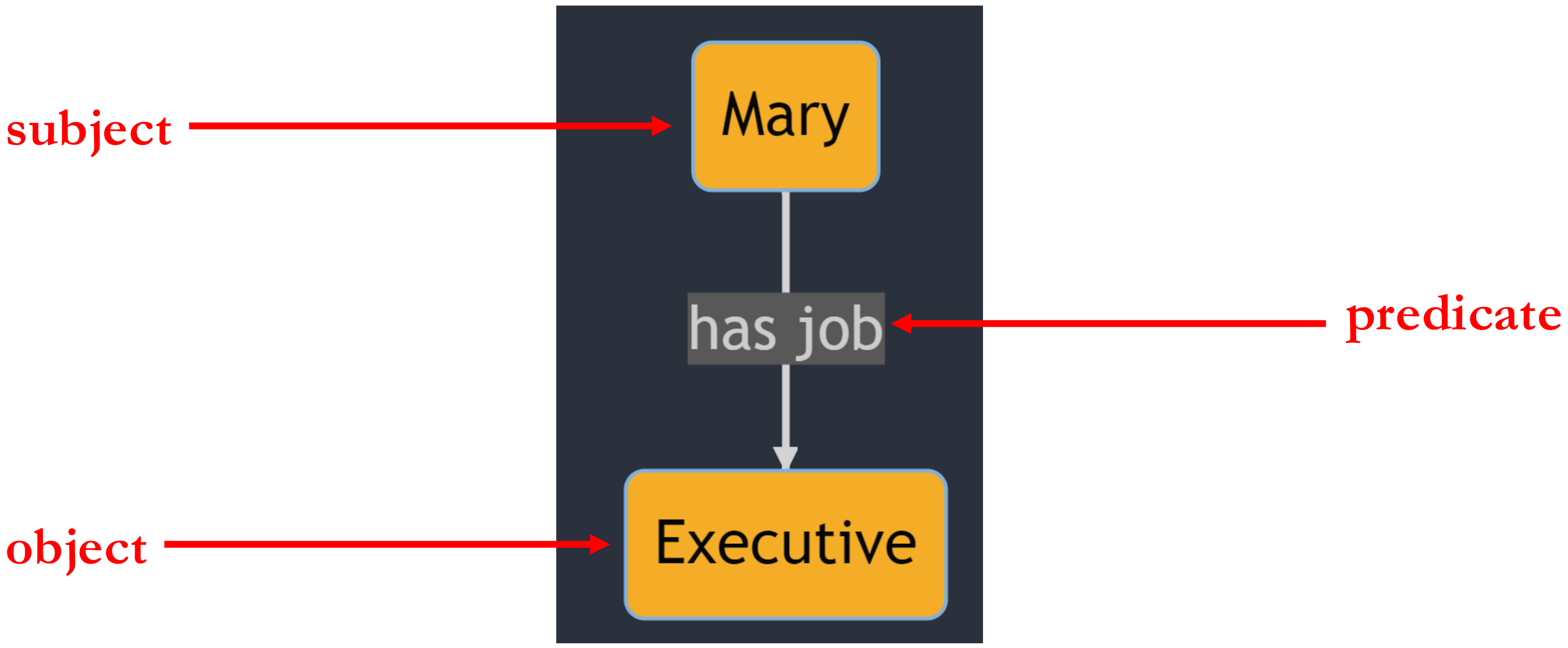
- RDF is:
  - A *data model*
  - That is based on *triples*
  - Which provide semantics for *distributed web data*



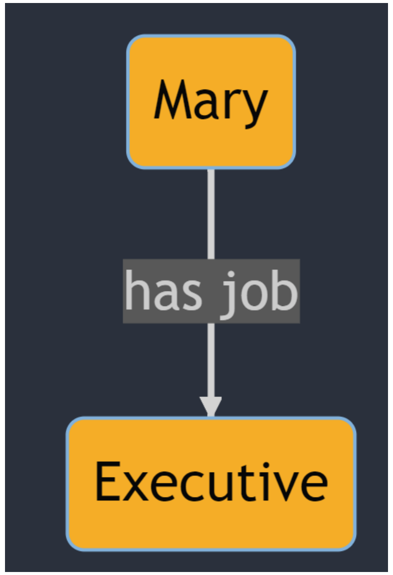
Triple Example



Triple Example



RDF Triple Structure



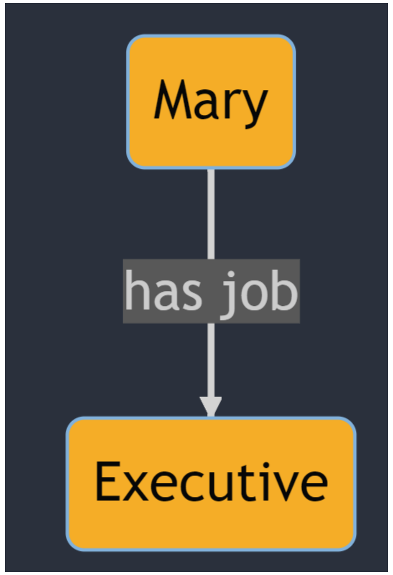
<https://www.example.com/occupation/Mary> <https://www.example.com/occupation/has_job> <https://www.example.com/occupation/Executive>

Subject

Predicate

Object

RDF Triple Structure



Uniform Resource Identifier (URI)

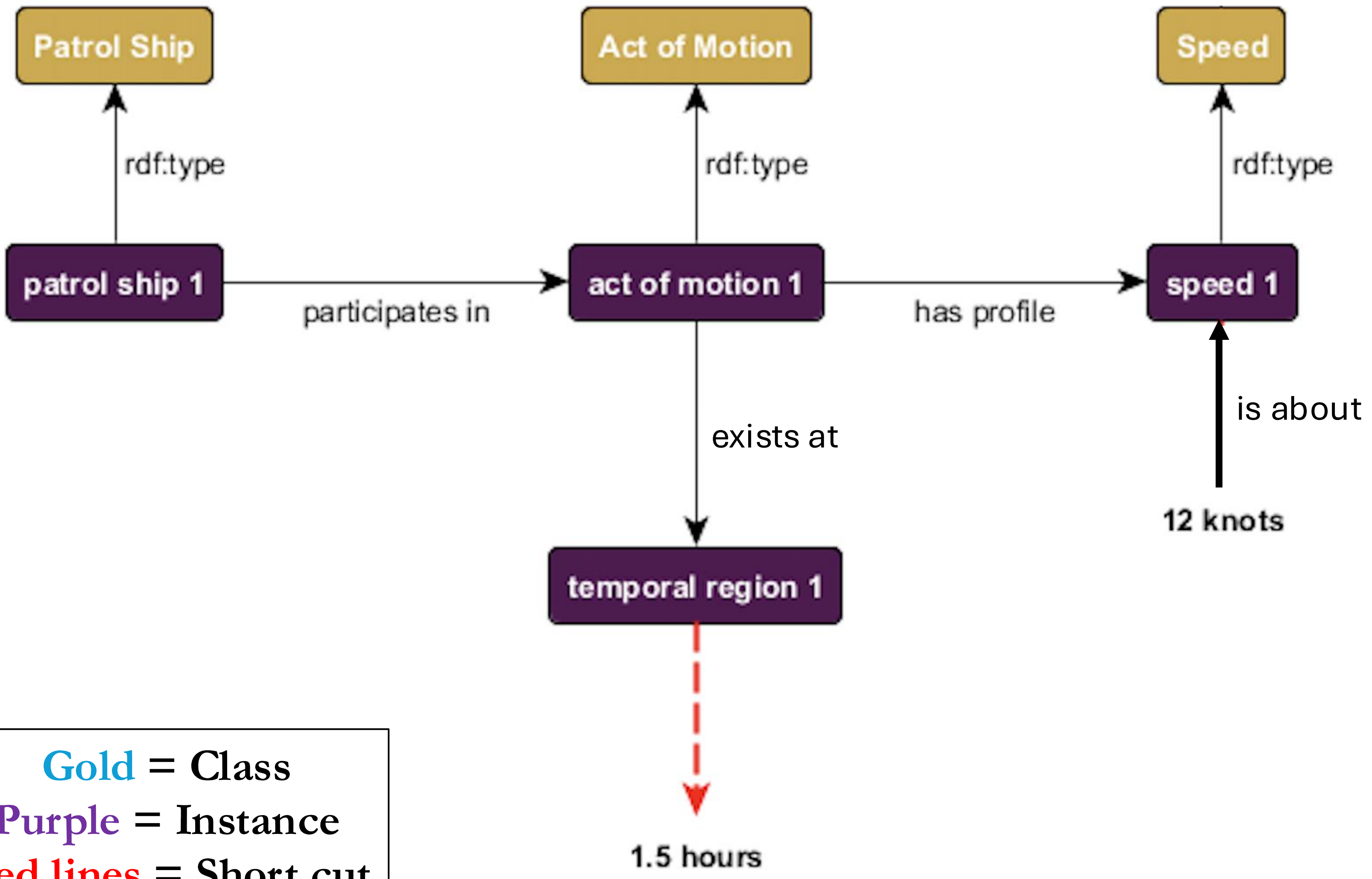
<https://www.example.com/occupation/Mary> <https://www.example.com/occupation/has_job> <https://www.example.com/occupation/Executive>

Subject

Predicate

Object

```
ex:patrol_ship_01  ex:participates_in  ex:act_of_motion_01 .
```



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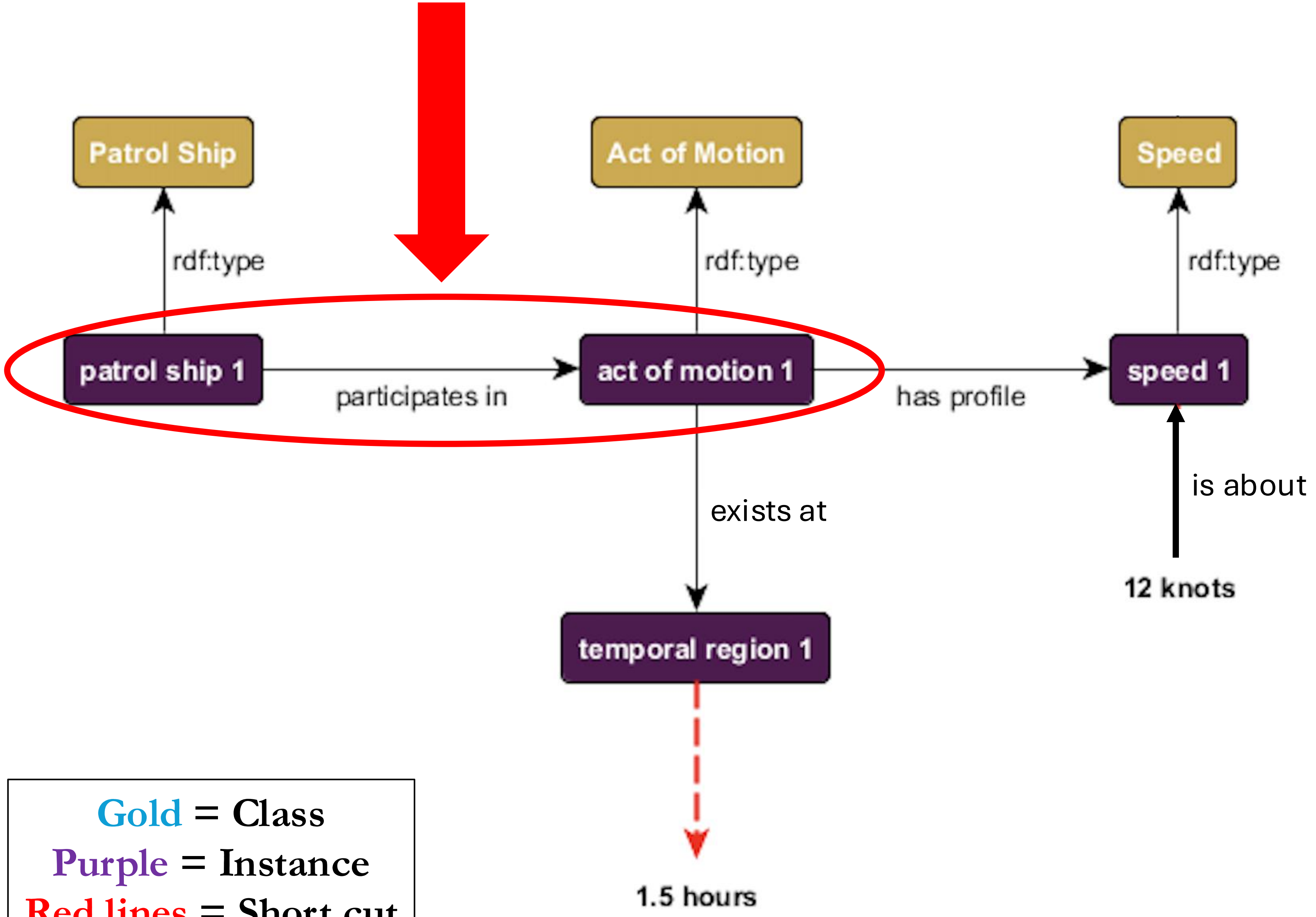

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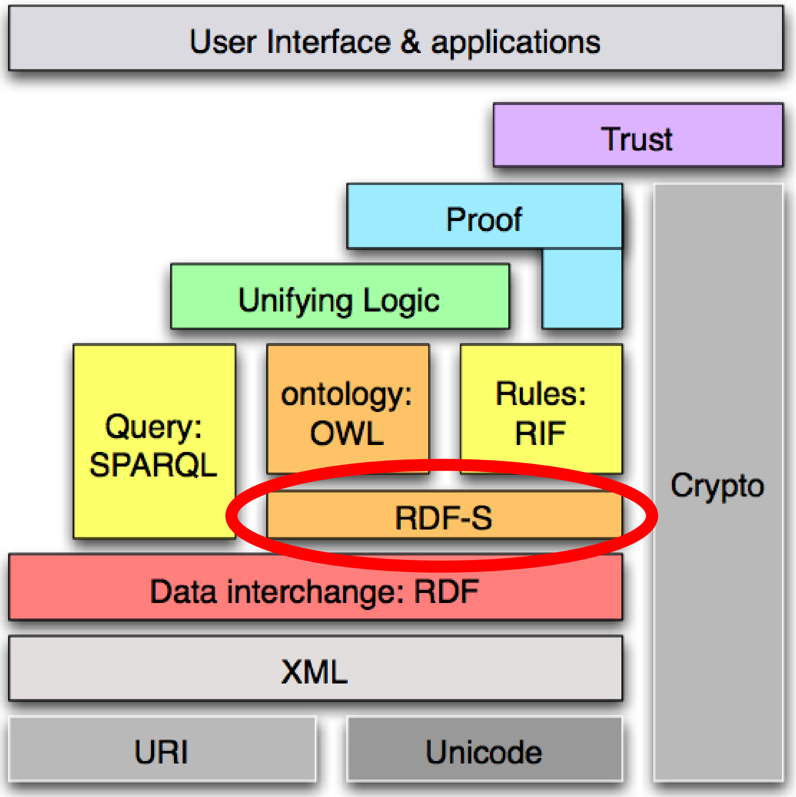
Semantic Web Stack

- The "S" in RDFS stands for:
  Schema – A syntax and semantics for an intended extension of the RDF syntax

- RDFS is:
  A *vocabulary*
  That is based on the *RDF data model*
  That extends the *RDF vocabulary*
  And provide semantics for *connecting RDF resources*



Resource Description Framework Schema

- RDFs is an extension of RDF facilitating representation of:

rdfs:Class
rdfs:subClassOf
rdfs:domain
rdfs:range
rdfs:subPropertyOf
rdfs:Resource

rdfs:Datatype
rdfs:Literal
rdfs:label
rdfs:comment
rdfs:isDefinedBy

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We often say X rdf:type Y and intend Y to be a class, but strictly speaking there is no notion of "class" in rdf
rdfs:Class introduces the syntax needed for such assertions

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rdfs:subClassOf facilitates representing the "is a" hierarchy
e.g. bfo:object rdfs:subClassOf bfo:MaterialEntity

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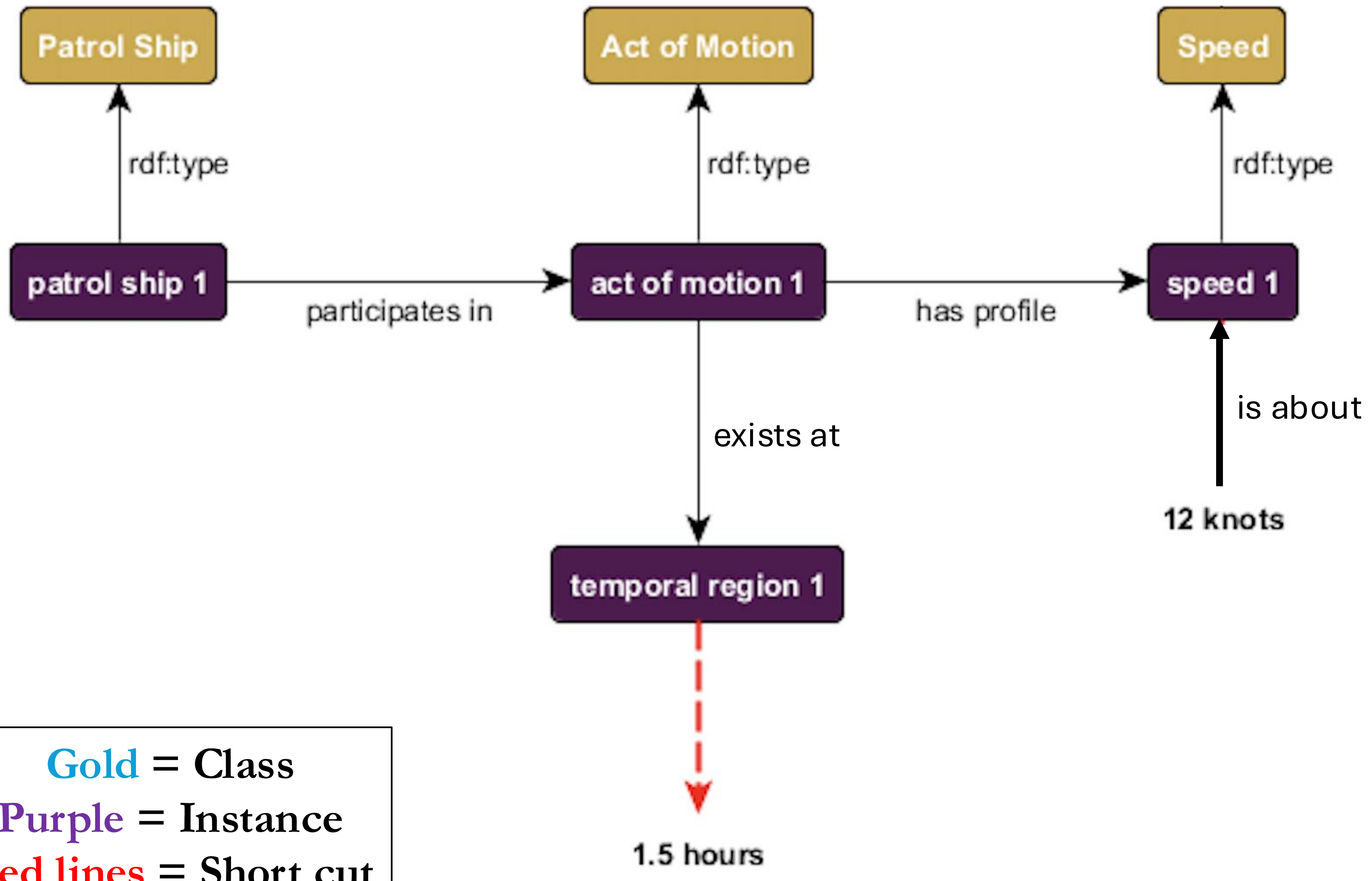
rdfs:comment
rdfs:isDefinedBy

rdfs:Datatype is a subclass of rdfs:Literal and is the class of all datatypes, such as "integer" or "decimal"

```
ex:PatrolShip    rdf:type    rdfs:Class .
ex:ActOfMotion  rdf:subClassOf   bfo:Process .
```



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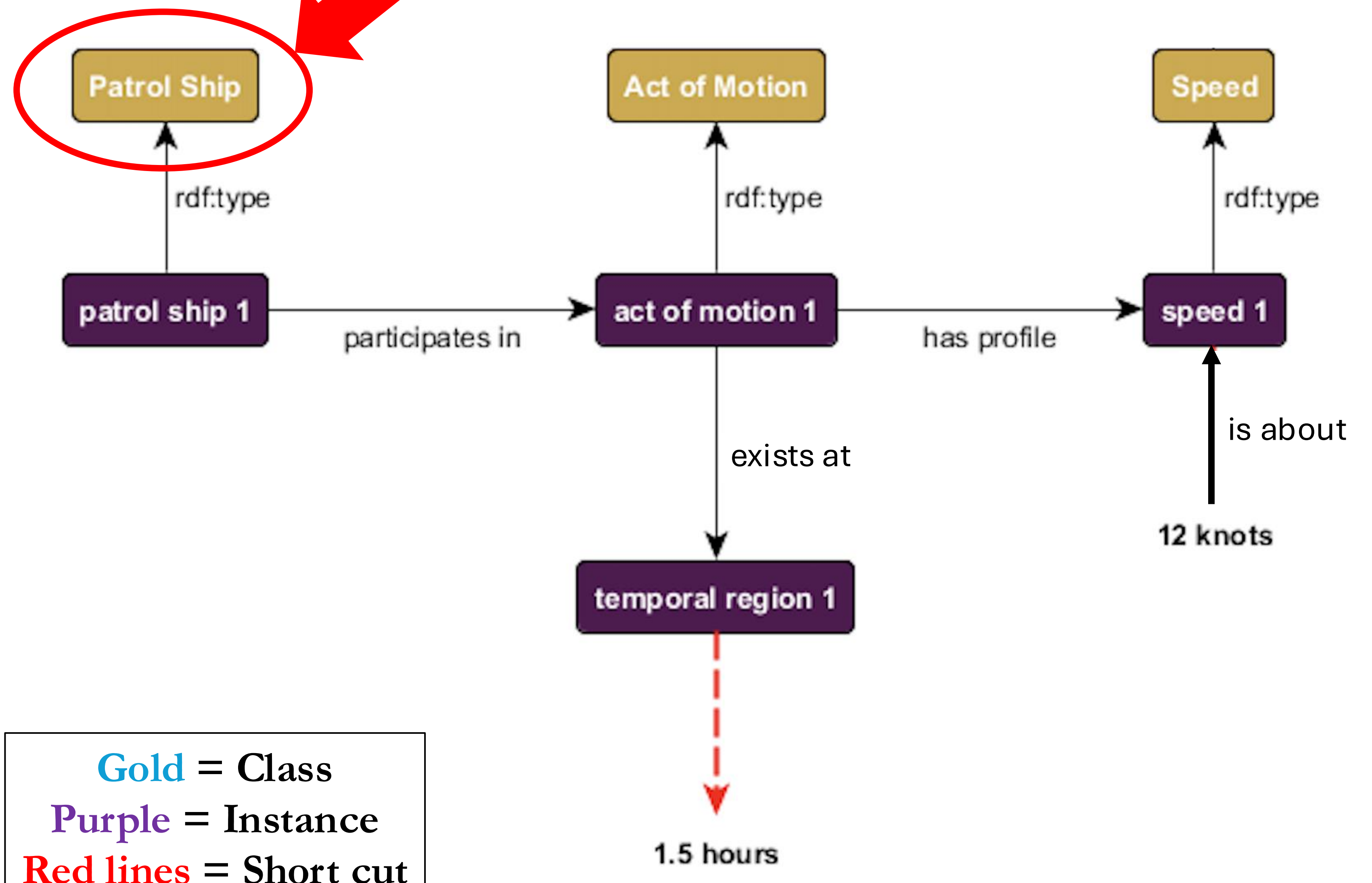

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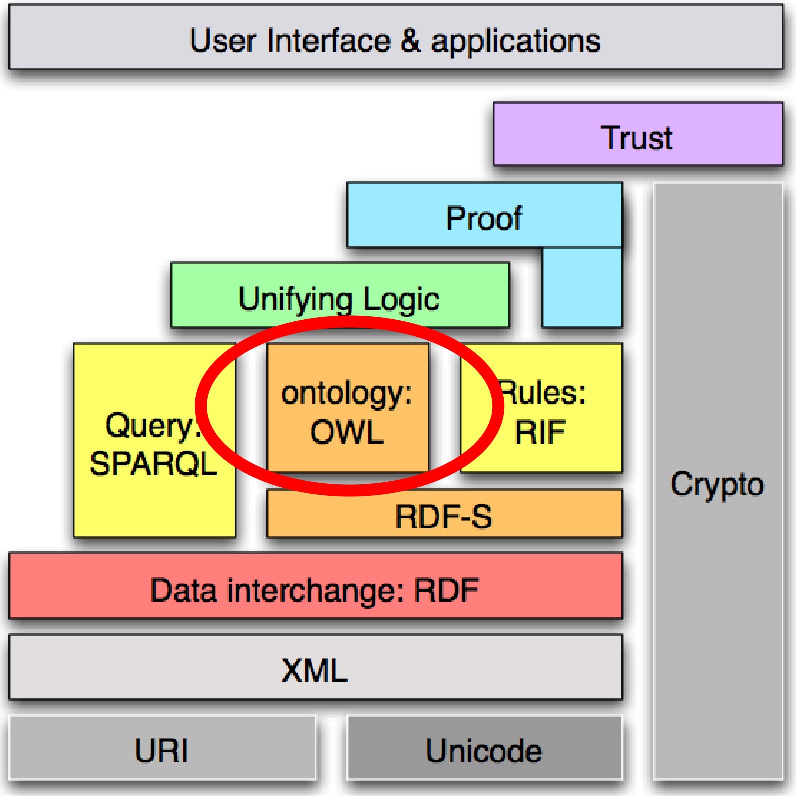
Semantic Web Stack

- "OWL" stands for:
  Web
  Ontology
  Language

- OWL is:
  A *family of vocabularies*
  That extend *RDF and RDFS*
  Which provide semantics for constructing *logical relationships among resources*



OWL2

- We will focus on the most widely used OWL language, called “OWL2”
- OWL2 has a clear connection to description logics
- The vocabulary includes terms for nothing, everything, union, etc. but also complex combinations such as disjointness
- Additionally, identity, saying something is related to something else, something is not related to something else, and something is an individual

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Why Encode in OWL?

Asserted Facts | **OWL Definition** | **Reasoner** | **Inferred Facts**

Asserted

```
patrol_boat_01 Type PatrolBoat
patrol_boat_01 participates_in motion_01
motion_01 Type ActOfMotion
```

Inferred

```
patrol_boat_01 Type MovingPatrolBoat
```